

lated enzyme, glutamine synthetase is not very active. Removal of AMP groups produces more active deadenylylated glutamine synthetase, and glutamine is formed. The glutamine synthetase system differs from the phosphorylase system in two ways: (1) AMP is used as the modifying agent, and (2) the modified form of glutamine synthetase is less active. Glutamine synthetase also is allosterically regulated. [Glutamine synthetase and its role in nitrogen metabolism \(pp. 211–12\)](#)

There are some advantages to using covalent modification for the regulation of enzyme activity. These interconvertible enzymes often are also allosteric. Because each form can respond differently to allosteric effectors, systems of covalently modified enzymes are able to respond to more stimuli in varied and sophisticated ways. Regulation can also be exerted on the enzymes that catalyze the covalent modifications, which adds a second level of regulation to the system.

Feedback Inhibition

The rate of many metabolic pathways is adjusted through control of the activity of the regulatory enzymes described in the preceding section. Every pathway has at least one **pacemaker enzyme** that catalyzes the slowest or rate-limiting reaction in the pathway. Because other reactions proceed more rapidly than the pacemaker reaction, changes in the activity of this enzyme directly alter the speed with which a pathway operates. Usually the first step in a pathway is a pacemaker reaction catalyzed by a regulatory enzyme. The end product of the pathway often inhibits this regulatory enzyme, a process known as **feedback inhibition** or **end product inhibition**. Feedback inhibition ensures balanced production of a pathway end product. If the end product becomes too concentrated, it inhibits the regulatory enzyme and slows its own synthesis. As the end product concentration decreases, pathway activity again increases and more product is formed. In this way feedback inhibition automatically matches end product supply with the demand. The previously discussed *E. coli* aspartate carbamoyltransferase is an excellent example of end product or feedback inhibition.

Frequently a biosynthetic pathway branches to form more than one end product. In such a situation the synthesis of pathway end products must be coordinated precisely. It would not do to have one end product present in excess while another is lacking. Branching biosynthetic pathways usually achieve a balance between end products through the use of regulatory enzymes at branch points (**figure 8.27**). If an end product is present in excess, it often inhibits the branch-point enzyme on the sequence leading to its formation, in this way regulating its own formation without affecting the synthesis of other products. In figure 8.27 notice that both products also inhibit the initial enzyme in the pathway. An excess of one product slows the flow of carbon into the whole pathway while inhibiting the appropriate branch-point enzyme. Because less carbon is required when a branch is not functioning, feedback inhibition of the initial pacemaker enzyme

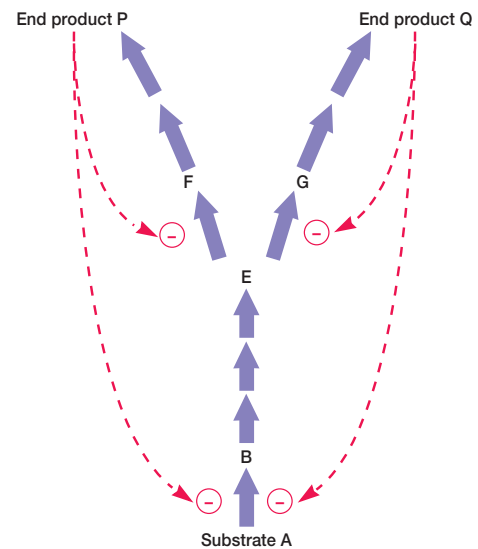


Figure 8.27 Feedback Inhibition. Feedback inhibition in a branching pathway with two end products. The branch-point enzymes, those catalyzing the conversion of intermediate E to F and G, are regulated by feedback inhibition. Products P and Q also inhibit the initial reaction in the pathway. A colored line with a minus sign at one end indicates that an end product, P or Q, is inhibiting the enzyme catalyzing the step next to the minus. See text for further explanation.

helps match the supply with the demand in branching pathways. The regulation of multiple branched pathways is often made even more sophisticated by the presence of **isoenzymes**, different enzymes that catalyze the same reaction. The initial pacemaker step may be catalyzed by several isoenzymes, each under separate and independent control. In such a situation an excess of a single end product reduces pathway activity but does not completely block pathway function because some isoenzymes are still active.

1. Define the following: allosteric enzyme, effector or modulator, and $[S]_{0.5}$ or $K_{0.5}$.
2. How can regulatory enzymes be influenced by reversible covalent modification? What groups are used for this purpose with glycogen phosphorylase and glutamine synthetase, and which forms of these enzymes are active?
3. What is a pacemaker enzyme? Feedback inhibition? How does feedback inhibition automatically adjust the concentration of a pathway end product? What are isoenzymes and why are they important in pathway regulation?

Summary

1. Energy is the capacity to do work. Living cells carry out three major kinds of work: chemical work of biosynthesis, transport work, and mechanical work.
2. The ultimate source of energy for most microbes is sunlight trapped by autotrophs and used to form organic material from CO₂. Photoautotrophs are then consumed by chemoheterotrophs.
3. ATP is the major energy currency and connects energy-generating processes with energy-using processes (figure 8.3).
4. The first law of thermodynamics states that energy is neither created nor destroyed.
5. The second law of thermodynamics states that changes occur in such a way that the randomness or disorder of the universe increases to the maximum possible. That is, entropy always increases during spontaneous processes.
6. The first and second laws can be combined to determine the amount of energy made available for useful work.

$$\Delta G = \Delta H - T\Delta S$$

In this equation the change in free energy (ΔG) is the energy made available for useful work, the change in enthalpy (ΔH) is the change in heat content, and the change in entropy is ΔS .

7. The standard free energy change (ΔG°) for a chemical reaction is directly related to the equilibrium constant.

8. In exergonic reactions ΔG° is negative and the equilibrium constant is greater than one; the reaction goes to completion as written. Endergonic reactions have a positive ΔG° and an equilibrium constant less than one (figure 8.5).
9. In oxidation-reduction (redox) reactions, electrons move from a donor, the reducing agent or reductant, to an acceptor, the oxidizing agent or oxidant. The standard reduction potential measures the tendency of the reducing agent to give up electrons.
10. Redox couples with more negative reduction potentials donate electrons to those with more positive potentials, and energy is made available during the transfer (figure 8.7).
11. Some most important electron carriers in cells are NAD⁺, NADP⁺, FAD, FMN, coenzyme Q, cytochromes, and the nonheme iron proteins.
12. Enzymes are protein catalysts that catalyze specific reactions.
13. Enzymes consist of a protein component, the apoenzyme, and often a nonprotein cofactor that may be a prosthetic group, a coenzyme, or a metal activator.
14. Enzymes speed reactions by binding substrates at their active sites and lowering the activation energy (figure 8.14).
15. The rate of an enzyme-catalyzed reaction increases with substrate concentration at low substrate levels and reaches a plateau (the maximum velocity) at saturating substrate concentrations. The Michaelis constant is the

- substrate concentration that the enzyme requires to achieve half maximal velocity (figure 8.17).
16. Enzymes have pH and temperature optima for activity.
17. Enzyme activity can be slowed by competitive and noncompetitive inhibitors.
18. The regulation of metabolism keeps cell components in proper balance and conserves metabolic energy and material.
19. The localization of metabolites and enzymes in different parts of the cell, called metabolic channeling, influences pathway activity. A common channeling mechanism is compartmentation.
20. Regulatory enzymes are usually allosteric enzymes, enzymes in which an effector or modulator binds reversibly to a regulatory site separate from the catalytic site and causes a conformational change in the enzyme to alter its activity (figure 8.21).
21. Aspartate carbamoyltransferase is an allosteric enzyme that is inhibited by CTP and activated by ATP.
22. Enzyme activity also can be regulated by reversible covalent modification. Two examples of such regulation are glycogen phosphorylase (phosphate addition) and glutamine synthetase (AMP addition).
23. The first enzyme in a pathway and enzymes at branch points often are subject to feedback inhibition by one or more end products. Excess end product slows its own synthesis (figure 8.27).

Key Terms

activation energy 162
active site 162
adenosine diphosphate (ADP) 155
adenosine 5'-triphosphate (ATP) 155
aerobic respiration 154
allosteric enzymes 165
apoenzyme 161
calorie 155
catalyst 161
catalytic site 162
chemical work 154
coenzyme 161
coenzyme Q or CoQ (ubiquinone) 159
cofactor 161
compartmentation 165
competitive inhibitor 164
cytochrome 159
denaturation 163
effector or modulator 165
end product inhibition 169
endergonic reaction 156
energy 154

enthalpy 156
entropy 156
enzyme 161
enzyme-substrate complex 162
equilibrium 156
equilibrium constant (K_{eq}) 156
exergonic reaction 156
feedback inhibition 169
ferredoxin 159
first law of thermodynamics 155
flavin adenine dinucleotide (FAD) 159
flavin mononucleotide (FMN) 159
free energy change 156
high-energy molecule 157
holoenzyme 161
isoenzymes 169
joule 155
mechanical work 154
metabolic channeling 165
Michaelis constant (K_m) 163
nicotinamide adenine dinucleotide (NAD⁺) 157

nicotinamide adenine dinucleotide phosphate (NADP⁺) 158
noncompetitive inhibitor 164
nonheme iron protein 159
oxidation-reduction (redox) reaction 157
oxidizing agent (oxidant) 157
pacemaker enzyme 169
phosphate group transfer potential 157
photosynthesis 154
product 161
prosthetic group 161
reducing agent (reductant) 157
regulatory site 165
reversible covalent modification 167
second law of thermodynamics 156
standard free energy change 156
standard reduction potential 157
substrate 161
thermodynamics 155
transition-state complex 162
transport work 154

Questions for Thought and Review

- Describe in general terms how energy from sunlight is spread throughout the biosphere.
- What sources of energy, other than sunlight, are used by microorganisms?
- Under what conditions would it be possible to create more order in a system without violating the second law of thermodynamics?
- Do living cells increase randomness or entropy within themselves? In the environment?
- Suppose that a chemical reaction had a large negative ΔG° value. What would this indicate about its equilibrium constant? If displaced from equilibrium, would it proceed rapidly to completion? Would much or little free energy be made available?
- Will electrons ordinarily move in an electron transport chain from cytochrome *c* ($E'_0 = +210$ mV) to O_2 ($E'_0 = +820$ mV) or in the opposite direction?
- If a person had a niacin deficiency, what metabolic process might well be adversely affected? Why?
- Draw a diagram showing how enzymes catalyze reactions by altering the activation energy. What is a transition-state complex? Use the diagram to discuss why enzymes do not change the equilibria of the reactions they catalyze.
- What special properties might an enzyme isolated from a psychrophilic bacterium have? Will enzymes need to lower the activation energy more or less in thermophiles than in psychrophiles?
- How might a substrate be able to regulate the activity of the enzyme using it?
- Describe how *E. coli* aspartate carbamoyltransferase is regulated, both in terms of the effects of modulators and the mechanism by which they exert their influence.
- What is the significance of the fact that regulatory enzymes often are located at pathway branch points?

Critical Thinking Questions

- How could electron transport be driven in the opposite direction? Why would it be desirable to do this?
- Take a look at the structures of macromolecules (appendix I). Which type has the most electrons to donate? Why are carbohydrates usually the primary source of electrons for nonautotrophic bacteria?
- Most enzymes do not operate at their biochemical optima inside cells. Why not?

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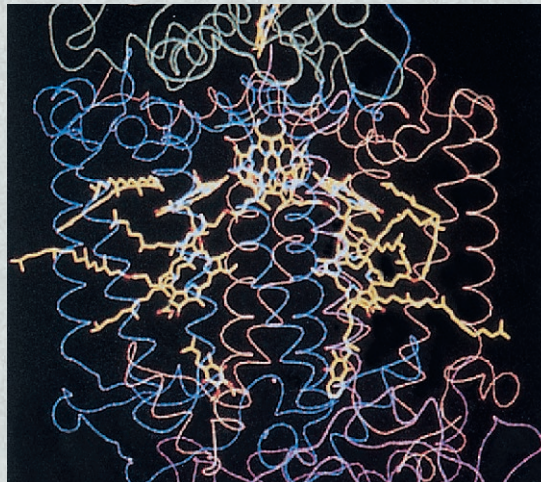
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CHAPTER 9

Metabolism: Energy Release and Conservation



The reaction center of the purple nonsulfur bacterium, *Rhodospseudomonas viridis*, with the bacteriochlorophylls and other prosthetic groups in yellow. These pigments trap light during photosynthesis.

Outline

- | | |
|--|--|
| 9.1 An Overview of Metabolism 173 | 9.6 Anaerobic Respiration 190 |
| 9.2 The Breakdown of Glucose to Pyruvate 176 | 9.7 Catabolism of Carbohydrates and Intracellular Reserve Polymers 191 |
| The Glycolytic Pathway 176 | Carbohydrates 191 |
| The Pentose Phosphate Pathway 177 | Reserve Polymers 192 |
| The Entner-Doudoroff Pathway 179 | 9.8 Lipid Catabolism 192 |
| 9.3 Fermentations 179 | 9.9 Protein and Amino Acid Catabolism 192 |
| 9.4 The Tricarboxylic Acid Cycle 183 | 9.10 Oxidation of Inorganic Molecules 193 |
| 9.5 Electron Transport and Oxidative Phosphorylation 184 | 9.11 Photosynthesis 195 |
| The Electron Transport Chain 184 | The Light Reaction in Eucaryotes and Cyanobacteria 196 |
| Oxidative Phosphorylation 187 | The Light Reaction in Green and Purple Bacteria 199 |
| The Yield of ATP in Glycolysis and Aerobic Respiration 189 | |

Concepts

1. Metabolism, the sum total of all chemical reactions occurring in the cell, can be divided into catabolism and anabolism. In catabolism, molecules are reduced in complexity and free energy is made available. Energy also can be derived from sunlight (photosynthesis) and the oxidation of inorganic nutrients. Anabolism involves the use of free energy to increase the complexity of molecules.
2. During catabolism, nutrients are funneled into a few common pathways for more efficient use of enzymes (a few pathways process a wide variety of nutrients).
3. The tricarboxylic acid cycle is the final pathway for the aerobic oxidation of nutrients to CO₂.
4. The majority of energy released in catabolism is generated by the movement of electrons from electron transport carriers with more negative reduction potentials to ones with more positive reduction potentials. Thus aerobic respiration is much more efficient than anaerobic catabolism.
5. A wide variety of electron acceptors can be used in catabolism: endogenous organic molecules (fermentation), O₂ (aerobic respiration), and oxidized exogenous inorganic and organic molecules other than O₂ (anaerobic respiration). Furthermore, reduced inorganic molecules as well as organic molecules can serve as electron donors for electron transport and ATP synthesis. Microbial catabolism is unique in the diversity of nutrients and mechanisms employed to make energy available.
6. In photosynthesis trapped light energy boosts electrons to more negative reduction potentials or higher energy levels. These energized electrons are then used to make ATP and NADPH or NADH during electron transport.
7. Proton motive force is generated by oxidative reactions and photosynthesis; it is used to power the production of ATP and other processes such as transport and bacterial motility.

It is in the fueling reactions that bacteria display their extraordinary metabolic diversity and versatility. Bacteria have evolved to thrive in almost all natural environments, regardless of the nature of available sources of carbon, energy, and reducing power. . . . The collective metabolic capacities of bacteria allow them to metabolize virtually every organic compound on this planet. . . .

—E. C. Neidhardt,

J. L. Ingraham, and M. Schaechter

Chapter 8 introduces the basic principles of thermodynamics, the energy cycle and the use of ATP as an energy currency, the nature and function of enzymes, and regulation of enzyme activity. With this background material in hand, microbial metabolism can be discussed. **Metabolism** is the total of all chemical reactions occurring in the cell. The flow of energy and the participation of enzymes make metabolism possible.

This chapter begins with an overview of metabolism. An introduction to carbohydrate degradation and fermentation follows. Next the formation of ATP driven by aerobic and anaerobic respiration is described. Then the breakdown of organic substances other than carbohydrates (i.e., lipids, proteins, and amino acids) is briefly surveyed. The chapter concludes with sections on the oxidation of inorganic molecules and the trapping of light energy by photosynthetic light reactions.

9.1 An Overview of Metabolism

Metabolism may be divided into two major parts. In **catabolism** [Greek *cata*, down, and *ballein*, to throw] larger and more complex molecules are broken down into smaller, simpler molecules with the release of energy. Some of this energy is trapped and made available for work; the remainder is released as heat. The trapped energy can then be used in anabolism, the second area of metabolism. **Anabolism** [Greek *ana*, up] is the synthesis of complex molecules from simpler ones with the input of energy. An anabolic process uses energy to increase the order of a system (*see chapter 8*).

Although the division of metabolism into two major parts is convenient and commonly employed, not all energy-yielding processes are comfortably encompassed by the previous definition of catabolism unless it is expanded to include processes that do not involve the degradation of complex organic molecules. In a broader sense, microorganisms usually use one of three sources of energy. Phototrophs capture radiant energy from the sun (**figure 9.1**). Chemoorganotrophs oxidize organic molecules to liberate energy, while chemolithotrophs employ inorganic nutrients as energy sources. All three processes will be discussed in this chapter, beginning with chemoorganotrophy. [Nutritional types of microorganisms \(pp. 97–98\)](#)

Microorganisms vary not only in their energy sources, but also in the electron acceptors used by chemotrophs (**figure 9.2**). Three major kinds of acceptors are employed. In **fermentation** [Latin *fermentare*, to cause to rise or ferment] the energy substrate is oxidized and degraded without the participation of an exogenous or externally derived electron acceptor. Usually the catabolic pathway produces an intermediate such as pyruvate that acts as the electron acceptor. Fermentation normally occurs under anaerobic conditions, but also occurs sometimes when oxygen is present. Of course, energy-yielding metabolism can make use of exogenous or externally derived electron acceptors. This metabolic process is called **respiration** and may be divided into two different types. In **aerobic respiration**, the final electron acceptor is oxygen, whereas the acceptor in **anaerobic respiration** is a different exogenous acceptor. Most often the acceptor in anaerobic respiration is inorganic (e.g., NO₃[−], SO₄^{2−}, CO₂, Fe³⁺, SeO₄^{2−}, and many others), but organic acceptors such as fumarate may be used. Most respiration involves the activity of an electron transport chain. The amount of available energy is quite different for fermentation and respiration. The electron acceptor in fermentation is at the same oxidation state as the original nutrient and there is no overall net oxidation of the nutrient. Thus only a limited amount of energy is made available. The acceptor in respiratory processes has reduction potential much more positive than the substrate and thus considerably more energy will be released during respiration (*see pp. 157–59 and figure 8.7*). In both aerobic and anaerobic respiration, ATP is formed as a result of electron transport chain activity. Electrons for the chain can be obtained from inorganic nutrients, and it is possible to derive energy from the oxidation of inorganic molecules rather than from organic nutrients. This ability is confined to a small group of prokaryotes called chemolithotrophs as mentioned previously (*see sections 22.1 and 22.2*).

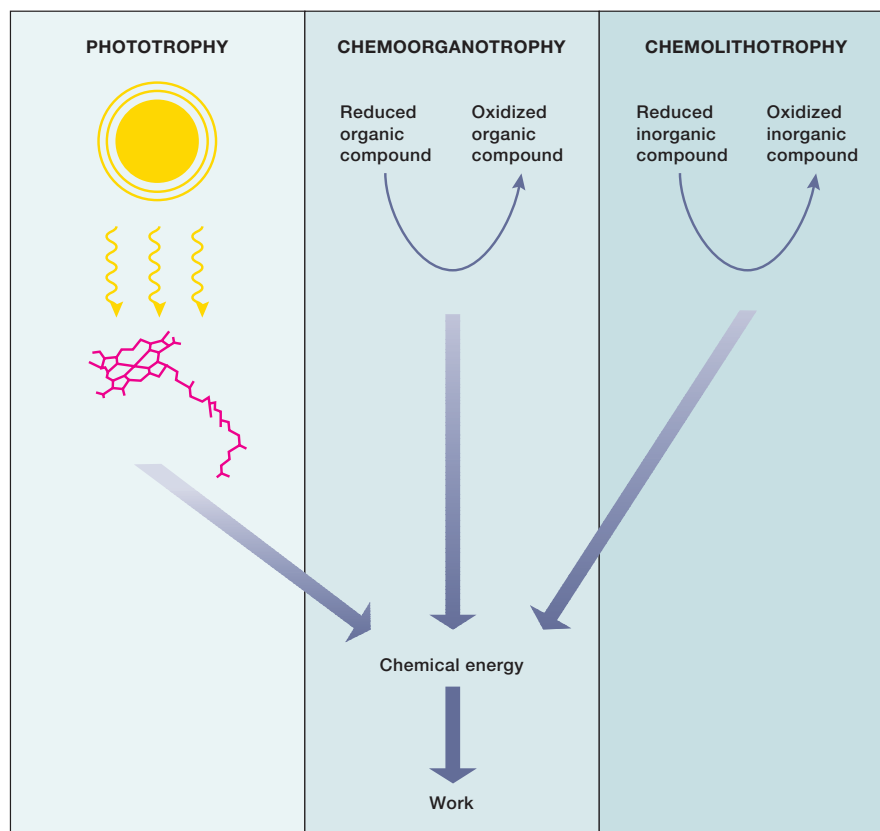


Figure 9.1 Sources of Energy for Microorganisms. Most microorganisms employ one of three energy sources. Phototrophs trap radiant energy from the sun using pigments such as bacteriochlorophyll and chlorophyll. Chemotrophs oxidize reduced organic and inorganic nutrients to liberate and trap energy. The chemical energy derived from these three sources can then be used in work as discussed in chapter 8.

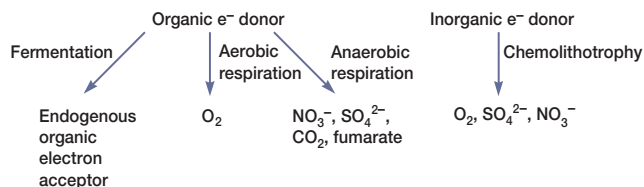


Figure 9.2 Patterns of Energy Release. Fermentation is the energy-yielding process in which an organic electron donor gives electrons to an endogenous acceptor, usually an intermediate derived from catabolism of the nutrient. In respiration, the electrons are donated to an exogenous acceptor, either oxygen (aerobic respiration) or some other acceptor such as nitrate or sulfate (anaerobic respiration). Reduced inorganic compounds also can serve as electron donors in energy production (chemolithotrophy).

It should be noted that these definitions of fermentation, aerobic respiration, and anaerobic respiration are slightly different from those often used by biologists and biochemists. Fermentation also may be defined as an energy-yielding process in which organic molecules serve as both electron donors and acceptors. Respiration is an energy-yielding process in which the acceptor is an inorganic molecule, either oxygen (aerobic respiration) or another inorganic acceptor (anaerobic respiration). Because microorganisms are so flexible and varied in their energy metabolism, the previous somewhat broader definitions will be used here.

Before learning about some of the more important catabolic pathways, it is best to look at the “lay of the land” and get our bearings. Albert Lehninger, a biochemist who worked at Johns Hopkins medical school, helped considerably in this task for chemoorganoheterotrophs by pointing out that aerobic metabolism may be divided into three stages (**figure 9.3**). In the first stage of catabolism, larger nutrient molecules (proteins, polysaccharides, and lipids) are hydrolyzed or otherwise broken

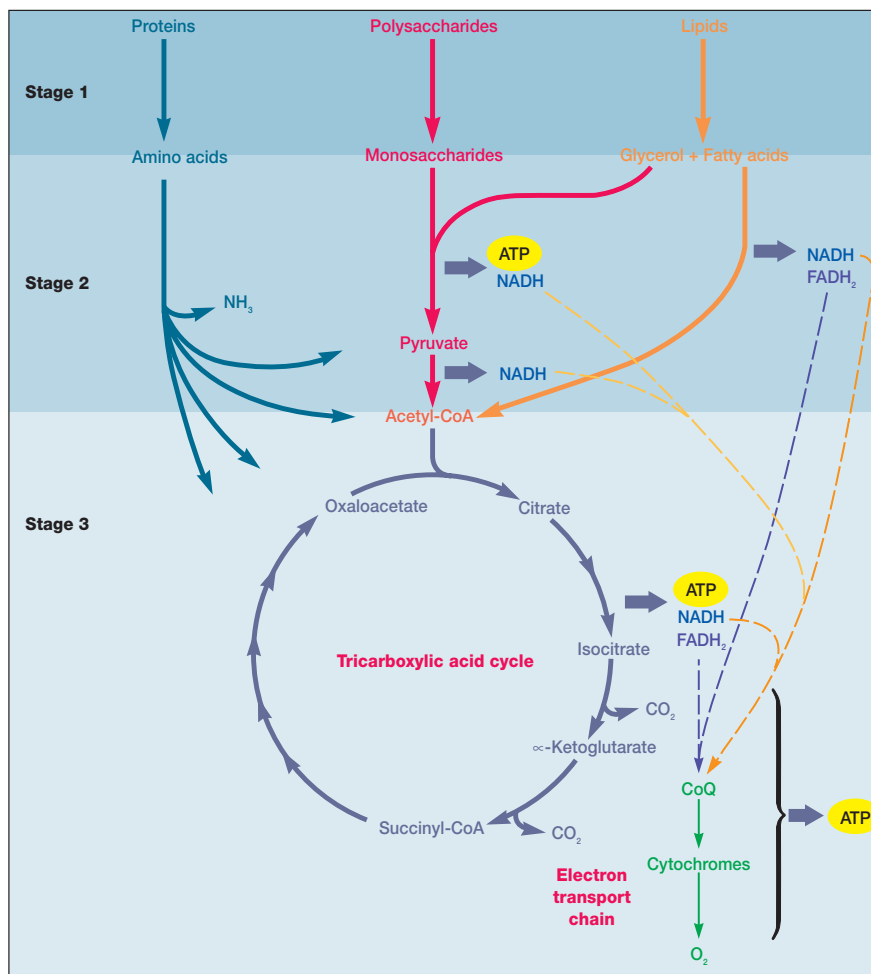


Figure 9.3 The Three Stages of Catabolism. A general diagram of aerobic catabolism in a chemorganoheterotroph showing the three stages in this process and the central position of the tricarboxylic acid cycle. Although there are many different proteins, polysaccharides, and lipids, they are degraded through the activity of a few common metabolic pathways. The dashed lines show the flow of electrons, carried by NADH and FADH₂, to the electron transport chain.

down into their constituent parts. The chemical reactions occurring during this stage do not release much energy. Amino acids, monosaccharides, fatty acids, glycerol, and other products of the first stage are degraded to a few simpler molecules in the second stage. Usually metabolites like acetyl coenzyme A, pyruvate, and tricarboxylic acid cycle intermediates are formed. The second-stage process can operate either aerobically or anaerobically and often produces some ATP as well as NADH and/or FADH₂. Finally, nutrient carbon is fed into the tricarboxylic acid cycle during the third stage of catabolism, and molecules are oxidized completely to CO₂ with the production of ATP, NADH, and FADH₂. The cycle operates aerobically and is

responsible for the release of much energy. Much of the ATP derived from the tricarboxylic acid cycle (and stage-two reactions) comes from the oxidation of NADH and FADH₂ by the electron transport chain. Oxygen, or sometimes another inorganic molecule, is the final electron acceptor.

Although this picture is somewhat oversimplified, it is useful in discerning the general pattern of catabolism. Notice that the microorganism begins with a wide variety of molecules and reduces their number and diversity at each stage. That is, nutrient molecules are funneled into ever fewer metabolic intermediates until they are finally fed into the tricarboxylic acid cycle. A common pathway often degrades many similar molecules (e.g., several

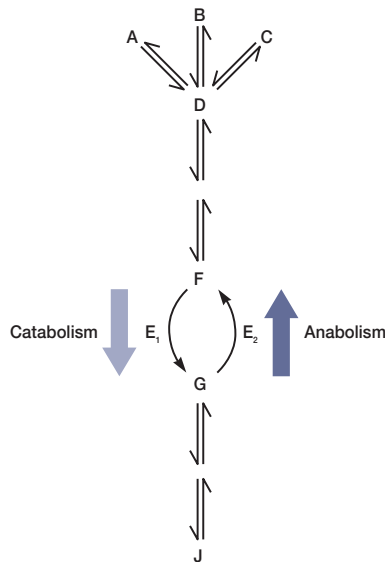


Figure 9.4 Amphibolic Pathway. A simplified diagram of an amphibolic pathway such as the glycolytic pathway. Note that the interconversion of intermediates F and G is catalyzed by two separate enzymes: E_1 operating in the catabolic direction and E_2 in the anabolic.

different sugars). These metabolic pathways consist of enzyme-catalyzed reactions arranged so that the product of one reaction serves as a substrate for the next. The existence of a few common catabolic pathways, each degrading many nutrients, greatly increases metabolic efficiency by avoiding the need for a large number of less metabolically flexible pathways. It is in the catabolic phase that microorganisms exhibit their nutritional diversity. Most microbial biosynthetic pathways closely resemble their counterparts in higher organisms. The uniqueness of microbial metabolism lies in the diversity of sources from which it generates ATP and NADH (figures 9.1 and 9.2).

Carbohydrates and other nutrients serve two functions in the metabolism of heterotrophic microorganisms: (1) they are oxidized to release energy, and (2) they supply carbon or building blocks for the synthesis of new cell constituents. Although many anabolic pathways are separate from catabolic routes, there are **amphibolic pathways** [Greek *amphi*, on both sides] that function both catabolically and anabolically. Two of the most important are the glycolytic pathway and the tricarboxylic acid cycle. Most reactions in these two pathways are freely reversible and can be used to synthesize and degrade molecules. The few irreversible catabolic steps are bypassed in biosynthesis with special enzymes that catalyze the reverse reaction (**figure 9.4**). For example, the enzyme fructose biphosphatase reverses the phosphofructokinase step when glucose is synthesized from pyruvate (*see p. 209*). The presence of two separate enzymes, one catalyzing the reversal of the other's reaction, permits independent regulation of the catabolic and anabolic functions of these amphibolic pathways.

1. Describe metabolism. How are catabolism and anabolism organized? What three major sources of energy are used by microorganisms? Define phototroph, chemoorganotroph, and chemolithotroph.
2. What kinds of electron acceptors do microorganisms use? Define fermentation, respiration, aerobic respiration, and anaerobic respiration.
3. Why are common pathways useful? What is an amphibolic pathway?

9.2 The Breakdown of Glucose to Pyruvate

Microorganisms employ several metabolic pathways to catabolize glucose and other sugars. Because of this metabolic diversity, their metabolism is often confusing. To avoid confusion as much as possible, the ways in which microorganisms degrade sugars to pyruvate and similar intermediates are introduced by focusing on only three routes: (1) glycolysis, (2) the pentose phosphate pathway, and (3) the Entner-Doudoroff pathway. Next the pathways of aerobic and anaerobic pyruvate metabolism are described. For the sake of simplicity, the chemical structures of metabolic intermediates are not used in pathway diagrams. [Diagrams of the glycolytic pathway and other major catabolic pathways with the structures of intermediates and enzyme names \(appendix II\)](#)

The Glycolytic Pathway

The **Embden-Meyerhof** or **glycolytic pathway** is undoubtedly the most common pathway for glucose degradation to pyruvate in stage two of catabolism. It is found in all major groups of microorganisms and functions in the presence or absence of O_2 . **Glycolysis** [Greek *glyco*, sweet, and *lysis*, a loosening] is located in the cytoplasmic matrix of procaryotes and eucaryotes.

The pathway as a whole may be divided into two parts (**figure 9.5** and *appendix II*). In the initial six-carbon stage, glucose is phosphorylated twice and eventually converted to fructose 1,6-bisphosphate. Other sugars are often fed into the pathway by conversion to glucose 6-phosphate or fructose 6-phosphate. This preliminary stage does not yield energy; in fact, two ATP molecules are expended for each glucose. These initial steps “prime the pump” by adding phosphates to each end of the sugar. The phosphates will soon be used to make ATP.

The three-carbon stage of glycolysis begins when the enzyme fructose 1,6-bisphosphate aldolase catalyzes the cleavage of fructose 1,6-bisphosphate into two halves, each with a phosphate group. One of the products, glyceraldehyde 3-phosphate, is converted directly to pyruvate in a five-step process. Because the other product, dihydroxyacetone phosphate, can be easily changed to glyceraldehyde 3-phosphate, both halves of fructose 1,6-bisphosphate are used in the three-carbon stage. Glyceraldehyde 3-phosphate is first oxidized with NAD^+ as the electron acceptor, and a phosphate is simultaneously incorporated to give a high-energy molecule called 1,3-bisphosphoglycerate. The high-energy phosphate on carbon one is subsequently donated to ADP

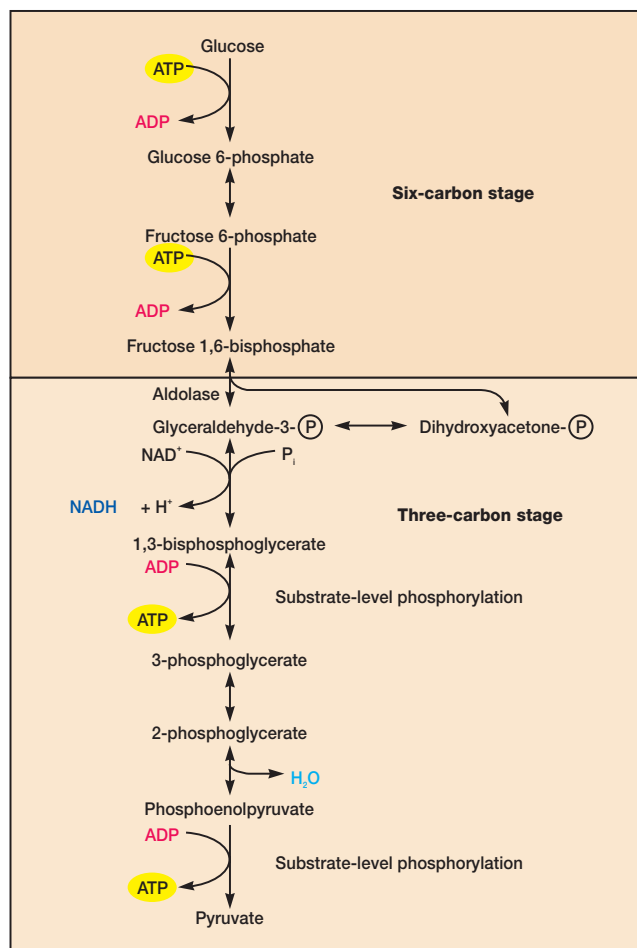


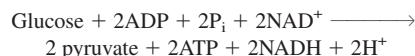
Figure 9.5 Glycolysis. The glycolytic pathway for the breakdown of glucose to pyruvate. The two stages of the pathway and their products are indicated.

to produce ATP. This synthesis of ATP is called **substrate-level phosphorylation** because ADP phosphorylation is coupled with the exergonic breakdown of a high-energy substrate molecule. ATP as an energy currency (pp. 155, 157)

A somewhat similar process generates a second ATP by substrate-level phosphorylation. The phosphate group on 3-phosphoglycerate shifts to carbon two, and 2-phosphoglycerate is dehydrated to form a second high-energy molecule, phosphoenolpyruvate. This molecule donates its phosphate to ADP forming a second ATP and pyruvate, the final product of the pathway.

The glycolytic pathway degrades one glucose to two pyruvates by the sequence of reactions just outlined. ATP and NADH are also produced. The yields of ATP and NADH may be calculated by considering the two stages separately. In the six-carbon stage two ATPs are used to form fructose 1,6-bisphosphate. For each glyceraldehyde 3-phosphate transformed into pyruvate, one NADH and two ATPs are formed. Because two glyceraldehyde 3-phosphates arise

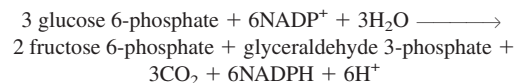
from a single glucose (one by way of dihydroxyacetone phosphate), the three-carbon stage generates four ATPs and two NADHs per glucose. Subtraction of the ATP used in the six-carbon stage from that produced in the three-carbon stage gives a net yield of two ATPs per glucose. Thus the catabolism of glucose to pyruvate in glycolysis can be represented by the following simple equation.



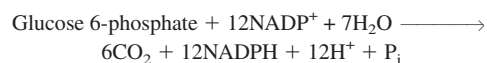
The Pentose Phosphate Pathway

A second pathway, the **pentose phosphate** or **hexose monophosphate pathway** may be used at the same time as the glycolytic pathway or the Entner-Doudoroff sequence. It can operate either aerobically or anaerobically and is important in biosynthesis as well as in catabolism.

The pentose phosphate pathway begins with the oxidation of glucose 6-phosphate to 6-phosphogluconate followed by the oxidation of 6-phosphogluconate to the pentose ribulose 5-phosphate and CO_2 (figure 9.6 and appendix II). NADPH is produced during these oxidations. Ribulose 5-phosphate is then converted to a mixture of three- through seven-carbon sugar phosphates. Two enzymes unique to this pathway play a central role in these transformations: (1) transketolase catalyzes the transfer of two-carbon ketol groups, and (2) transaldolase transfers a three-carbon group from sedoheptulose 7-phosphate to glyceraldehyde 3-phosphate (figure 9.7). The overall result is that three glucose 6-phosphates are converted to two fructose 6-phosphates, glyceraldehyde 3-phosphate, and three CO_2 molecules, as shown in the following equation.



These intermediates are used in two ways. The fructose 6-phosphate can be changed back to glucose 6-phosphate while glyceraldehyde 3-phosphate is converted to pyruvate by glycolytic enzymes. The glyceraldehyde 3-phosphate also may be returned to the pentose phosphate pathway through glucose 6-phosphate formation. This results in the complete degradation of glucose 6-phosphate to CO_2 and the production of a great deal of NADPH.



The pentose phosphate pathway has several catabolic and anabolic functions that are summarized as follows:

1. NADPH from the pentose phosphate pathway serves as a source of electrons for the reduction of molecules during biosynthesis.
2. The pathway synthesizes four- and five-carbon sugars for a variety of purposes. The four-carbon sugar erythrose 4-phosphate is used to synthesize aromatic amino acids and vitamin B_6 (pyridoxal). The pentose ribose 5-phosphate is a major component of nucleic acids, and ribulose 1,5-bisphosphate is the primary CO_2 acceptor in photosynthesis. Note that when a microorganism is growing

Figure 9.6 The Pentose Phosphate Pathway.

The conversion of three glucose 6-phosphate molecules to two fructose 6-phosphates and a glyceraldehyde 3-phosphate is traced. The fructose 6-phosphates are changed back to glucose 6-phosphate. The glyceraldehyde 3-phosphate can be converted to pyruvate or combined with a molecule of dihydroxyacetone phosphate (from the glyceraldehyde 3-phosphate formed by a second turn of the pathway) to yield fructose 6-phosphate.

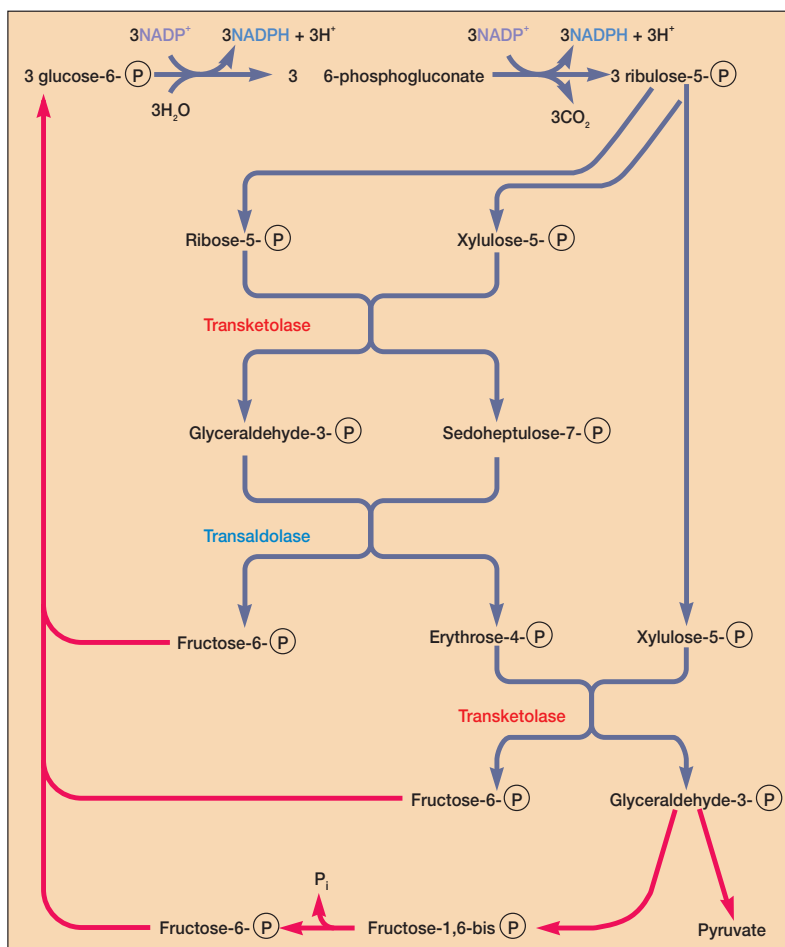
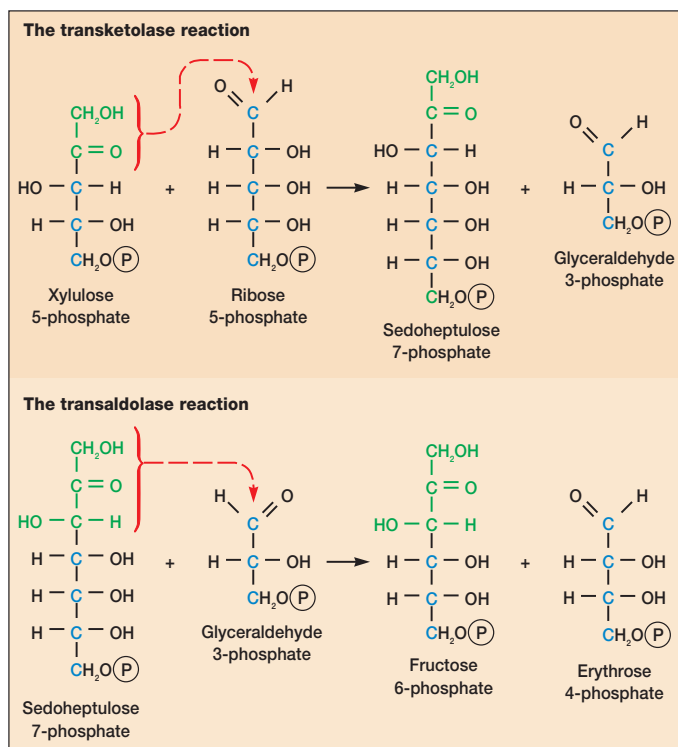


Figure 9.7 Transketolase and Transaldolase. Examples of the transketolase and transaldolase reactions of the pentose phosphate pathway. The groups transferred in these reactions are in color.



on a pentose carbon source, the pathway also can supply carbon for hexose production (e.g., glucose is needed for peptidoglycan synthesis).

- Intermediates in the pentose phosphate pathway may be used to produce ATP. Glyceraldehyde 3-phosphate from the pathway can enter the three-carbon stage of the glycolytic pathway and be converted to ATP and pyruvate. The latter may be oxidized in the tricarboxylic acid cycle to provide more energy. In addition, some NADPH can be converted to NADH, which yields ATP when it is oxidized by the electron transport chain. Because five-carbon sugars are intermediates in the pathway, the pentose phosphate pathway can be used to catabolize pentoses as well as hexoses.

Although the pentose phosphate pathway may be a source of energy in many microorganisms, it is more often of greater importance in biosynthesis. Several functions of the pentose phosphate pathway are mentioned again in chapter 10 when biosynthesis is considered more directly.

The Entner-Doudoroff Pathway

Although the glycolytic pathway is the most common route for the conversion of hexoses to pyruvate, another pathway with a similar role has been discovered. The **Entner-Doudoroff pathway** begins with the same reactions as the pentose phosphate pathway, the formation of glucose 6-phosphate and 6-phosphogluconate (figure 9.8 and appendix II). Instead of being further oxidized, 6-phosphogluconate is dehydrated to form 2-keto-3-deoxy-6-phosphogluconate or KDPG, the key intermediate in this pathway. KDPG is then cleaved by KDPG aldolase to pyruvate and glyceraldehyde 3-phosphate. The glyceraldehyde 3-phosphate is converted to pyruvate in the bottom portion of the glycolytic pathway. If the Entner-Doudoroff pathway degrades glucose to pyruvate in this way, it yields one ATP, one NADPH, and one NADH per glucose metabolized.

Most bacteria have the glycolytic and pentose phosphate pathways, but some substitute the Entner-Doudoroff pathway for glycolysis. The Entner-Doudoroff pathway is generally found in *Pseudomonas*, *Rhizobium*, *Azotobacter*, *Agrobacterium*, and a few other gram-negative genera. Very few gram-positive bacteria have this pathway, with *Enterococcus faecalis* being a rare exception.

- Summarize the major features of the glycolytic pathway, the pentose phosphate pathway, and the Entner-Doudoroff sequence. Include the starting points, the products of the pathways, the critical or unique enzymes, the ATP yields, and the metabolic roles each pathway has.
- What is substrate-level phosphorylation?

9.3 Fermentations

In the absence of aerobic or anaerobic respiration, NADH is not oxidized by the electron transport chain because no external electron acceptor is available. Yet NADH produced in the glycolytic pathway during the oxidation of glyceraldehyde 3-phosphate to

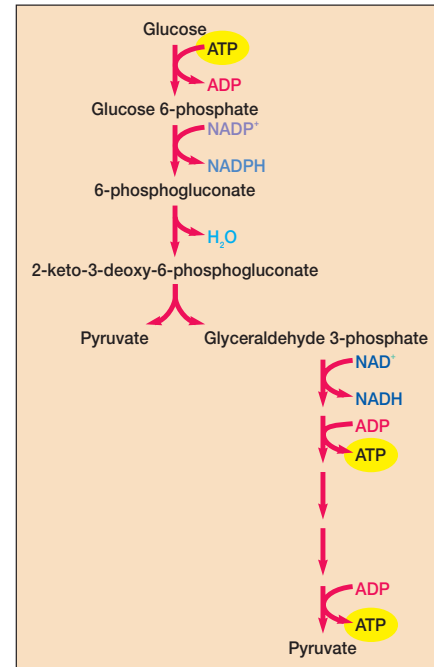


Figure 9.8 The Entner-Doudoroff Pathway. The sequence leading from glyceraldehyde 3-phosphate to pyruvate is catalyzed by enzymes common to the glycolytic pathway.

1,3-bisphosphoglycerate (figure 9.5) must still be oxidized back to NAD^+ . If NAD^+ is not regenerated, the oxidation of glyceraldehyde 3-phosphate will cease and glycolysis will stop. Many microorganisms solve this problem by slowing or stopping pyruvate dehydrogenase activity and using pyruvate or one of its derivatives as an electron and hydrogen acceptor for the reoxidation of NADH in a fermentation process (figure 9.9). This may lead to the production of more ATP. The process is so effective that some chemoorganoheterotrophs do not carry out respiration even when oxygen or another exogenous acceptor is available. There are many kinds of fermentations, and they often are characteristic of particular microbial groups (figure 9.10). A few more common fermentations are introduced here, and several others are discussed at later points. Two unifying themes should be kept in mind when microbial fermentations are examined: (1) NADH is oxidized to NAD^+ , and (2) the electron acceptor is often either pyruvate or a pyruvate derivative. In fermentation the substrate is partially oxidized, ATP is formed by substrate-level phosphorylation only, and oxygen is not needed.

Many fungi and some bacteria, algae, and protozoa ferment sugars to ethanol and CO_2 in a process called **alcoholic fermentation**. Pyruvate is decarboxylated to acetaldehyde, which is then reduced to ethanol by alcohol dehydrogenase with NADH as the electron donor (figure 9.10, number 2). **Lactic acid fermentation**, the reduction of pyruvate to lactate (figure 9.10, number 1), is even more common. It is present in bacteria (lactic acid bacteria, *Bacillus*), algae (*Chlorella*), some water molds, protozoa, and even in

Figure 9.9 Reoxidation of NADH During Fermentation. NADH from glycolysis is reoxidized by being used to reduce pyruvate or a pyruvate derivative (X). Either lactate or reduced product Y result.

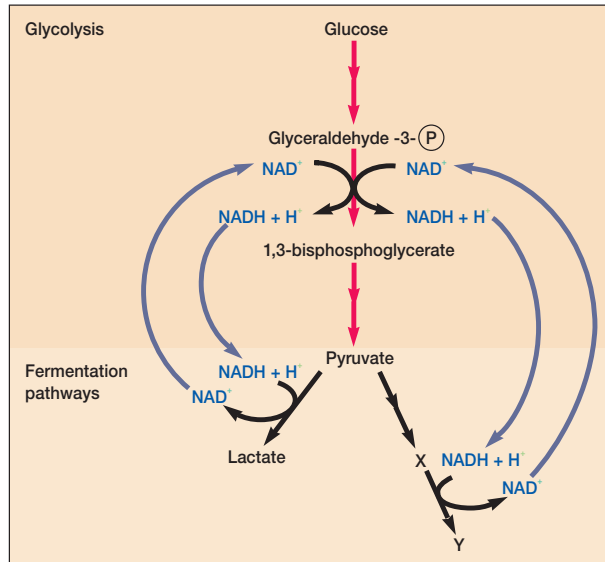


Figure 9.10 Some Common Microbial Fermentations. Only pyruvate fermentations are shown for the sake of simplicity; many other organic molecules can be fermented. Most of these pathways have been simplified by deletion of one or more steps and intermediates. Pyruvate and major end products are shown in color.

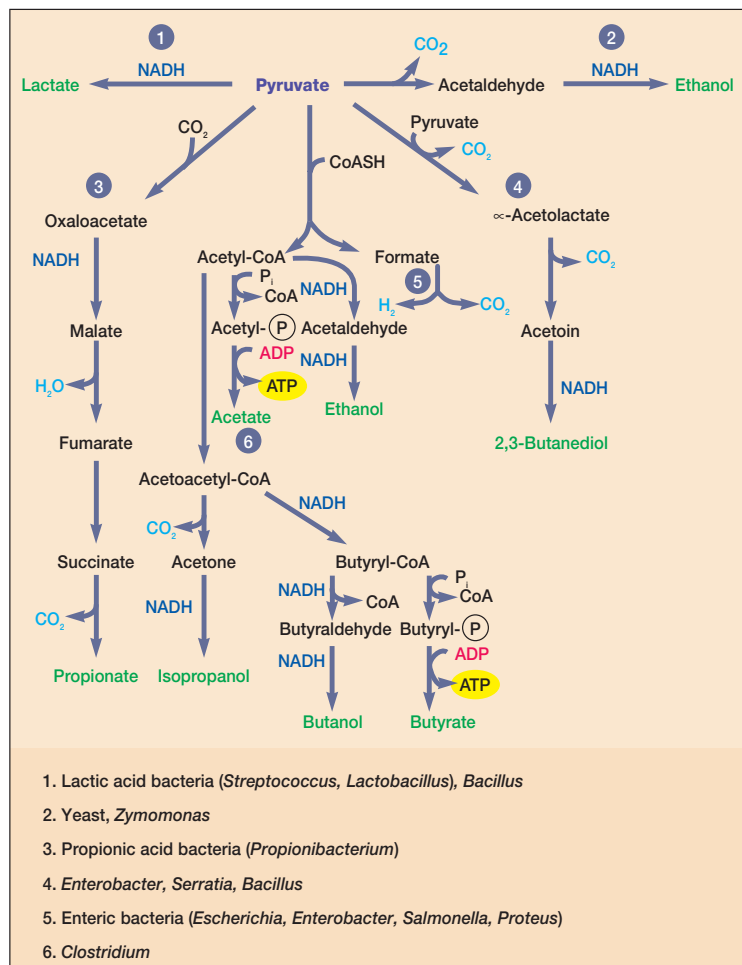


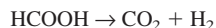
Table 9.1 Mixed Acid Fermentation Products of *Escherichia coli*

	Fermentation Balance (μM Product/100 μM Glucose)	
	Acid Growth (pH 6.0)	Alkaline Growth (pH 8.0)
Ethanol	50	50
Formic acid	2	86
Acetic acid	36	39
Lactic acid	80	70
Succinic acid	11	15
Carbon dioxide	88	2
Hydrogen gas	75	0.5
Butanediol	0	0

animal skeletal muscle. Lactic acid fermenters can be separated into two groups. **Homolactic fermenters** use the glycolytic pathway and directly reduce almost all their pyruvate to lactate with the enzyme lactate dehydrogenase. **Heterolactic fermenters** form substantial amounts of products other than lactate; many produce lactate, ethanol, and CO_2 by way of the phosphoketolase pathway (see section 23.4). The *Enterobacteriaceae* (pp. 505–7)

Alcoholic and lactic acid fermentations are quite useful. Alcoholic fermentation by yeasts produces alcoholic beverages; CO_2 from this fermentation causes bread to rise. Lactic acid fermentation can spoil foods, but also is used to make yogurt, sauerkraut, and pickles. The role of fermentations in food production is discussed in chapter 41.

Many bacteria, especially members of the family *Enterobacteriaceae*, can metabolize pyruvate to formic acid and other products in a process sometimes called the formic acid fermentation (figure 9.10, number 5). Formic acid may be converted to H_2 and CO_2 by formic hydrogenlyase (a combination of at least two enzymes).



There are two types of formic acid fermentation. **Mixed acid fermentation** results in the excretion of ethanol and a complex mixture of acids, particularly acetic, lactic, succinic, and formic acids (table 9.1). If formic hydrogenlyase is present, the formic acid will be degraded to H_2 and CO_2 . This pattern is seen in *Escherichia*, *Salmonella*, *Proteus*, and other genera. The second type, **butanediol fermentation**, is characteristic of *Enterobacter*, *Serratia*, *Erwinia*, and some species of *Bacillus* (figure 9.10, number 4). Pyruvate is converted to acetoin, which is then reduced to 2,3-butanediol with NADH. A large amount of ethanol is also produced, together with smaller amounts of the acids found in mixed acid fermentation.

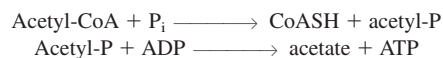
Formic acid fermentations are very useful in identification of members of the *Enterobacteriaceae*. Butanediol fermenters can be distinguished from mixed acid fermenters in three ways.

1. The Voges-Proskauer test is a colorimetric procedure that detects the acetoin precursor of butanediol (figure 9.10) and

is positive with butanediol fermenters but not with mixed acid fermenters. The Voges-Proskauer test is used by both the Enterotube II and API 20E microbial identification systems to identify enteric bacteria.

2. Mixed acid fermenters produce four times more acidic products than neutral ones, whereas butanediol fermenters form mainly neutral products. Thus mixed acid fermenters acidify incubation media to a much greater extent. This is the basis of the methyl red test. The test is positive only for mixed acid fermentation because the pH drops below 4.4 and the color of the indicator changes from yellow to red.
3. CO_2 and H_2 arise in equal amounts from formic hydrogenlyase activity during mixed acid fermentation. Butanediol fermenters produce excess CO_2 and the CO_2/H_2 ratio is closer to 5:1.

Formic acid fermenters sometimes generate ATP while reoxidizing NADH. They use acetyl-CoA to synthesize acetyl phosphate, which then donates its phosphate to ADP.



Microorganisms carry out fermentations other than those already mentioned (Box 9.1). Protozoa and fungi often ferment sugars to lactate, ethanol, glycerol, succinate, formate, acetate, butanediol, and additional products.

Substances other than sugars also are fermented by microorganisms. For example, some members of the genus *Clostridium* (see pp. 523–24) prefer to ferment mixtures of amino acids. Proteolytic clostridia such as the pathogens *C. sporogenes* and *C. botulinum* will carry out the **Stickland reaction** in which one amino acid is oxidized and a second amino acid acts as the electron acceptor. Figure 9.11 shows the way in which alanine is oxidized and glycine reduced to produce acetate, CO_2 , and NH_3 . Some ATP is formed from acetyl phosphate by substrate-level phosphorylation, and the fermentation is quite useful for growing in anaerobic, protein-rich environments. The Stickland reaction is used to oxidize several amino acids: alanine, leucine, isoleucine, valine, phenylalanine, tryptophan, and histidine. Bacteria also ferment amino acids (e.g., alanine, glycine, glutamate, threonine, and arginine) by other mechanisms. In addition to sugars and amino acids, organic acids such as acetate, lactate, propionate, and citrate are fermented. Some of these fermentations are of great practical importance. For example, citrate can be converted to diacetyl and give flavor to fermented milk (see pp. 978–79).

1. What are fermentations and why are they so useful to many microorganisms? Can ATP be produced during fermentation?
2. Briefly describe alcoholic, lactic acid, and formic acid fermentations. How do mixed acid fermenters and butanediol fermenters differ from each other?
3. What is the Stickland reaction?

Box 9.1

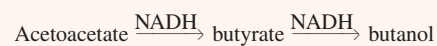
Microbiology and World War I

The unique economic pressures of wartime sometimes provide incentive for scientific discovery. Two examples from the First World War involve the production of organic solvents by the microbial fermentation of readily available carbohydrates, such as starch or molasses.

The German side needed glycerol to make nitroglycerin. At one time the Germans had imported their glycerol, but such imports were prevented by the British naval blockade. The German scientist Carl Neuberg knew that trace levels of glycerol were usually produced during the alcoholic fermentation of sugar by *Saccharomyces cerevisiae*. He sought to develop a modified fermentation in which the yeasts would produce glycerol instead of ethanol. Normally acetaldehyde is reduced to ethanol by NADH and alcohol dehydrogenase (figure 9.10, pathway 2). Neuberg found that this reaction could be prevented by the addition of 3.5% sodium sulfite at pH 7.0. The bisulfite ions reacted with acetaldehyde and made it unavailable for reduction to ethanol. Because the yeast cells still had to regenerate their NAD⁺ even though acetaldehyde was no longer available, Neuberg suspected that they would simply increase the rate of glycerol synthesis. Glycerol is normally produced by the reduction of dihydroxyacetone phosphate (a glycolytic intermediate) to glycerol phosphate with NADH, followed by the hydrolysis of glycerol phosphate to glycerol. Neuberg's hunch was correct, and German breweries were converted to glycerol manufacture by his procedure, eventually producing 1,000 tons of glycerol per month. Glycerol production by *S. cerevisiae* was not economically competitive under peacetime conditions and was ended. Today glycerol is produced microbially by the halophilic alga *Dunaliella salina*, in which high concentrations of intracellular glycerol accumulate to counterbalance the osmotic pressure from the high level

of extracellular salt. *Dunaliella* grows in habitats such as the Great Salt Lake of Utah and seaside rock pools.

The British side needed the organic solvents acetone and butanol. Butanol was required for the production of artificial rubber, whereas acetone was used as a solvent from nitrocellulose in the manufacture of the smokeless explosive powder cordite. Prior to 1914 acetone was made by the dry heating (pyrolysis) of wood. Between 80 and 100 tons of birch, beech, or maple wood were required to make 1 ton of acetone. When war broke out, the demand for acetone quickly exceeded the existing world supply. However, by 1915 Chaim Weizmann, a young Jewish scientist working in Manchester, England, had developed a fermentation process by which the anaerobic bacterium *Clostridium acetobutylicum* converted 100 tons of molasses or grain into 12 tons of acetone and 24 tons of butanol (most clostridial fermentations stop at butyric acid).



This time the British and Canadian breweries were converted until new fermentation facilities could be constructed. Weizmann improved the process by finding a convenient way to select high-solvent producing strains of *C. acetobutylicum*. Because the strains most efficient in these fermentations also made the most heat-resistant spores, Weizmann merely isolated the survivors from repeated 100°C heat shocks. Acetone and butanol were made commercially by this fermentation process until it was replaced by much cheaper petrochemicals in the late 1940s and 1950s. In 1948 Chaim Weizmann became the first president of the State of Israel.

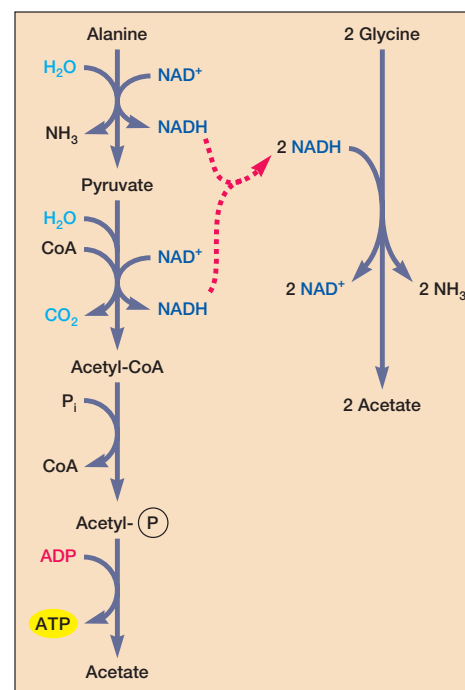


Figure 9.11 The Stickland Reaction. Alanine is oxidized to acetate and glycine is used to reoxidize the NADH generated during alanine degradation. The fermentation also produces some ATP.

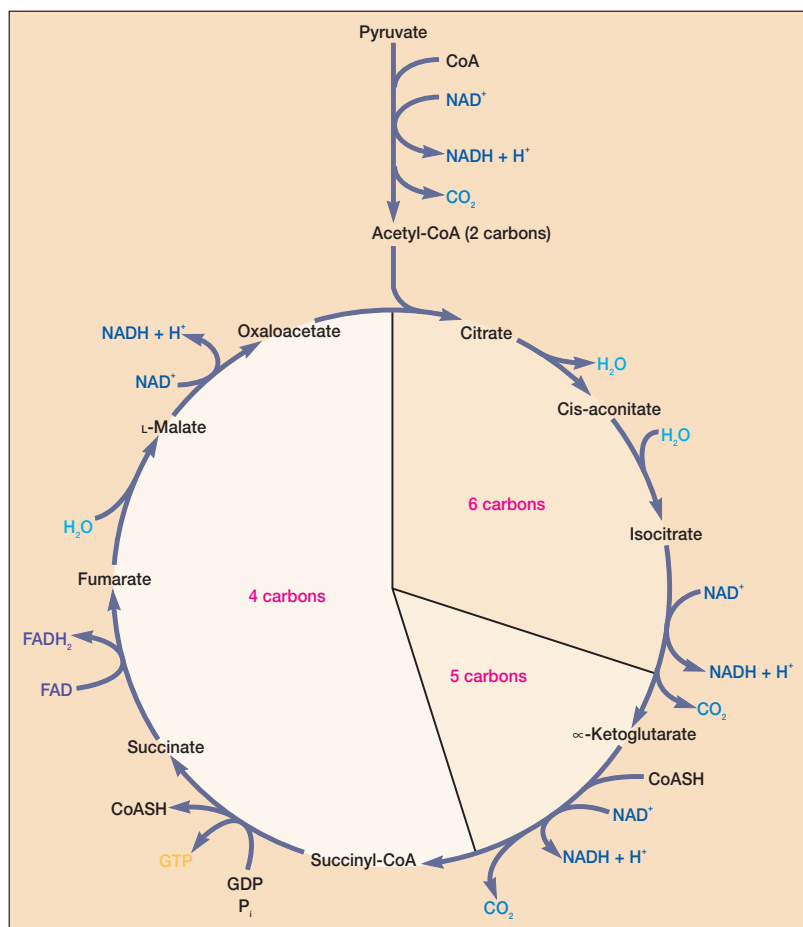


Figure 9.12 The Tricarboxylic Acid Cycle. The cycle may be divided into three stages based on the size of its intermediates. The three stages are separated from one another by two decarboxylation reactions (reactions in which carboxyl groups are lost as CO₂). The pyruvate dehydrogenase complex forms acetyl-CoA through pyruvate oxidation.

9.4 The Tricarboxylic Acid Cycle

Although some energy is obtained from the breakdown of glucose to pyruvate by the pathways previously described, much more is released when pyruvate is degraded aerobically to CO₂ in stage three of catabolism. The multienzyme system called the pyruvate dehydrogenase complex first oxidizes pyruvate to form CO₂ and **acetyl coenzyme A (acetyl-CoA)**, an energy-rich molecule composed of coenzyme A and acetic acid joined by a high-energy thiol ester bond (figure 9.12). Acetyl-CoA arises from the catabolism of many carbohydrates, lipids, and amino acids (figure 9.3). It can be further degraded in the tricarboxylic acid cycle.

The substrate for the **tricarboxylic acid (TCA) cycle, citric acid cycle, or Krebs cycle** is acetyl-CoA (figure 9.12 and appendix II). The traditional way to think about the cycle is in terms of its intermediates and products, and the chemistry involved in each step. In the first reaction acetyl-CoA is condensed with a four-carbon intermediate, oxaloacetate, to form citrate and to begin the six-carbon stage. Citrate (a tertiary alcohol) is rearranged to give isocitrate, a more readily oxidized secondary alcohol. Isocitrate is

subsequently oxidized and decarboxylated twice to yield α-ketoglutarate, then succinyl-CoA. At this point two NADHs are formed and two carbons are lost from the cycle as CO₂. Because two carbons were added as acetyl-CoA at the start, balance is maintained and no net carbon is lost. The cycle now enters the four-carbon stage during which two oxidation steps yield one FADH₂ and one NADH per acetyl-CoA. In addition, GTP (a high-energy molecule equivalent to ATP) is produced from succinyl-CoA by substrate-level phosphorylation. Eventually oxaloacetate is reformed and ready to join with another acetyl-CoA. Inspection of figure 9.12 shows that the TCA cycle generates two CO₂s, three NADHs, one FADH₂, and one GTP for each acetyl-CoA molecule oxidized.

Another way to think of the TCA cycle is in terms of its function as a pathway that oxidizes acetyl-CoA to CO₂. From this perspective, the first step is the attachment of an acetyl group to the acetyl carrier, oxaloacetate, to form citrate. The second stage begins with citrate and ends in the formation of succinyl-CoA. Here, the acetyl carrier portion of citrate loses two carbons when it is oxidized to give two CO₂s. The third and last stage converts

succinyl-CoA back to oxaloacetate, the acetyl carrier, so that it can pick up another acetyl group.

TCA cycle enzymes are widely distributed among microorganisms. The complete cycle appears to be functional in many aerobic bacteria, free-living protozoa, and most algae and fungi. This is not surprising because the cycle is such an important source of energy. However, the facultative anaerobe *E. coli* does not use the full TCA cycle under anaerobic conditions or when the glucose concentration is high but does at other times. Even those microorganisms that lack the complete TCA cycle usually have most of the cycle enzymes, because one of TCA cycle's major functions is to provide carbon skeletons for use in biosynthesis. [The role of the tricarboxylic acid cycle in biosynthesis \(pp. 214–16\)](#)

1. Give the substrate and products of the tricarboxylic acid cycle. Describe its organization in general terms. What are its two major functions?
2. What chemical intermediate links glycolysis to the TCA cycle?
3. In what eucaryotic organelle is the TCA cycle found? Where is the cycle located in procaryotes?

9.5 Electron Transport and Oxidative Phosphorylation

Little ATP has been synthesized up to this point. Only the equivalent of four ATP molecules is directly synthesized when one glucose is oxidized to six CO₂ molecules by way of glycolysis and the TCA cycle. Most ATP generated comes from the oxidation of NADH and FADH₂ in the electron transport chain. The mitochondrial electron transport chain will be examined first because it has been so well studied. Then we will turn to bacterial chains, and finish with a discussion of ATP synthesis.

The Electron Transport Chain

The mitochondrial **electron transport chain** is composed of a series of electron carriers that operate together to transfer electrons from donors, like NADH and FADH₂, to acceptors, such as O₂ (refer to diagram of a mitochondrial electron transport chain in [figure 9.13](#)). The electrons flow from carriers with more negative reduction potentials to those with more positive potentials and eventually combine with O₂ and H⁺ to form water. This pattern of electron flow is exactly the same as seen in the electron tower that was described in chapter 8 (*see figure 8.7*). The electrons move down this potential gradient much like water flowing down a series of rapids. The difference in reduction potentials between O₂ and NADH is large, about 1.14 volts, and makes possible the release of a great deal of energy. The potential changes at several points in the chain are large enough to provide sufficient energy for ATP production, much like the energy from waterfalls can be harnessed by waterwheels and used to generate electricity. The electron transport chain breaks up the large overall energy release into small steps. Some of the liberated energy is trapped in the form of ATP. As will be seen shortly, electron transport at these

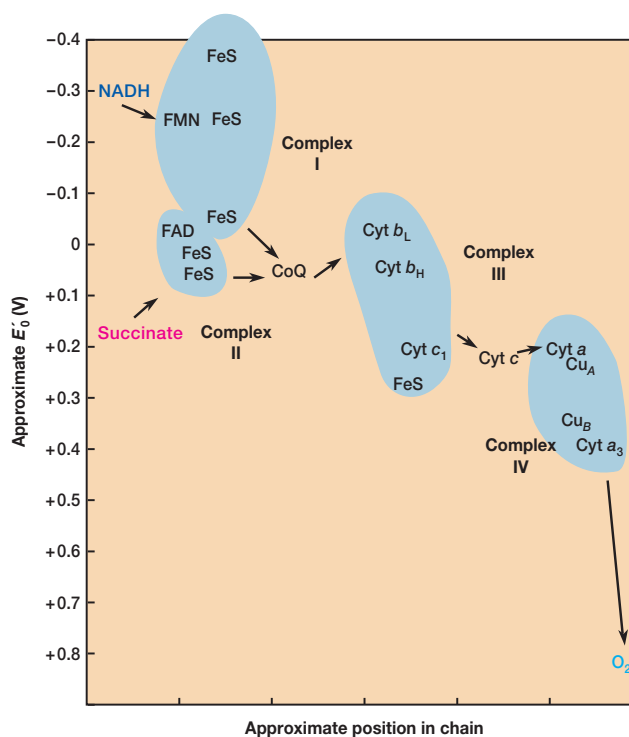


Figure 9.13 The Mitochondrial Electron Transport Chain. Many of the more important carriers are arranged at approximately the correct reduction potential and sequence. In the eucaryotic mitochondrion, they are organized into four complexes that are linked by coenzyme Q (CoQ) and cytochrome *c* (Cyt *c*). Electrons flow from NADH and succinate down the reduction potential gradient to oxygen. See text for details.

points may generate proton and electrical gradients. These gradients can then drive ATP synthesis.

The electron transport chain carriers reside within the inner membrane of the mitochondrion or in the bacterial plasma membrane. The mitochondrial system is arranged into four complexes of carriers, each capable of transporting electrons part of the way to O₂ ([figure 9.14](#)). Coenzyme Q and cytochrome *c* connect the complexes with each other.

The process by which energy from electron transport is used to make ATP is called **oxidative phosphorylation**. Thus as many as three ATP molecules may be synthesized from ADP and Pi when a pair of electrons pass from NADH to an atom of O₂. This is the same thing as saying that the phosphorus to oxygen (P/O) ratio is equal to 3. Because electrons from FADH₂ only pass two oxidative phosphorylation points, the maximum P/O ratio for FADH₂ is 2. The actual P/O ratios may be less than 3.0 and 2.0 in eucaryotic mitochondria.

The preceding discussion has focused on the eucaryotic mitochondrial electron transport chain. Although some bacterial chains resemble the mitochondrial chain, they are frequently very different. They vary in their electron carriers (e.g., in their cy-

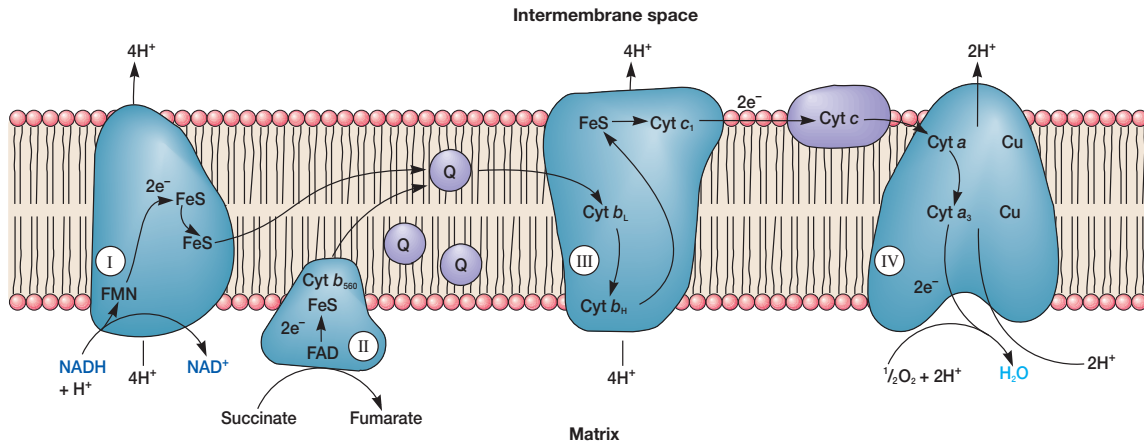
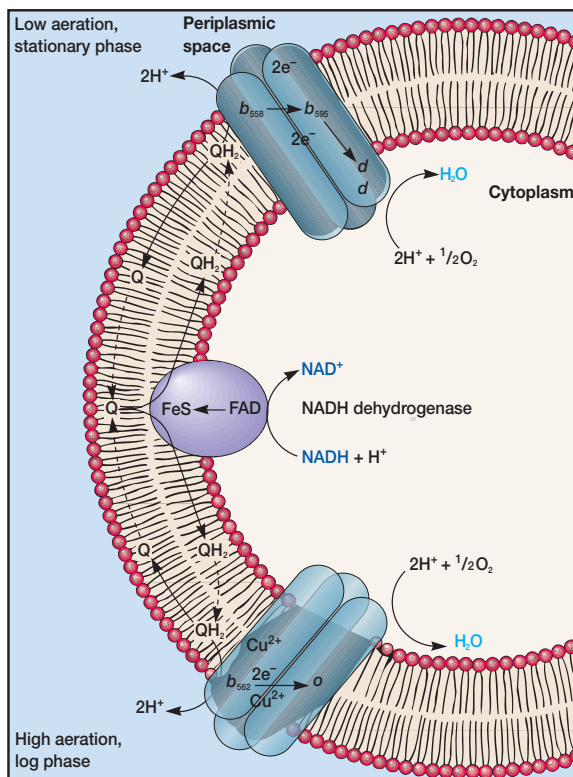


Figure 9.14 The Chemiosmotic Hypothesis Applied to Mitochondria. In this scheme the carriers are organized asymmetrically within the inner membrane so that protons are transported across as electrons move along the chain. Proton release into the intermembrane space occurs when electrons are transferred from carriers, such as FMN and coenzyme Q (Q), that carry both electrons and protons to components like nonheme iron proteins (FeS proteins) and cytochromes (Cyt) that transport only electrons. Complex IV pumps protons across the membrane as electrons pass from cytochrome *a* to oxygen. Coenzyme Q transports electrons from complexes I and II to complex III. Cytochrome *c* moves electrons between complexes III and IV. The number of protons moved across the membrane at each site per pair of electrons transported is still somewhat uncertain; the current consensus is that at least 10 protons must move outward during NADH oxidation.



tochromes) and may be extensively branched. Electrons often can enter at several points and leave through several terminal oxidases. Bacterial chains also may be shorter and have lower P/O ratios than mitochondrial transport chains. Thus procaryotic and eucaryotic electron transport chains differ in details of construction although they operate using the same fundamental principles.

The electron transport chains of *Escherichia coli* and *Paracoccus denitrificans* will serve as examples of these differences. A simplified view of the *E. coli* transport chain is shown in **figure 9.15**. Although it transports electrons from NADH to acceptors and moves protons across the plasma membrane, the *E. coli* chain is quite different from the mitochondrial chain. For example, it is branched and contains a quite different array of cytochromes. Coenzyme Q or ubiquinol donates electrons to both branches, but they operate under different growth conditions. The cytochrome *d* branch has very high affinity for oxygen and functions at low oxygen levels. It is not as efficient as the cytochrome *o* branch because it does not actively pump protons. The cytochrome *o* branch has moderately high affinity for oxygen, is a proton pump, and operates at higher oxygen concentrations.

Figure 9.15 The Aerobic Respiratory System of *E. coli*. NADH is the electron source. Ubiquinone-8 (Q) connects the NADH dehydrogenase with two terminal oxidase systems. The upper branch operates when the bacterium is in stationary phase and there is little oxygen. At least five cytochromes are involved: *b*₅₅₈, *b*₅₉₅, *b*₅₆₂, *d*, and *o*. The lower branch functions when *E. coli* is growing rapidly with good aeration.

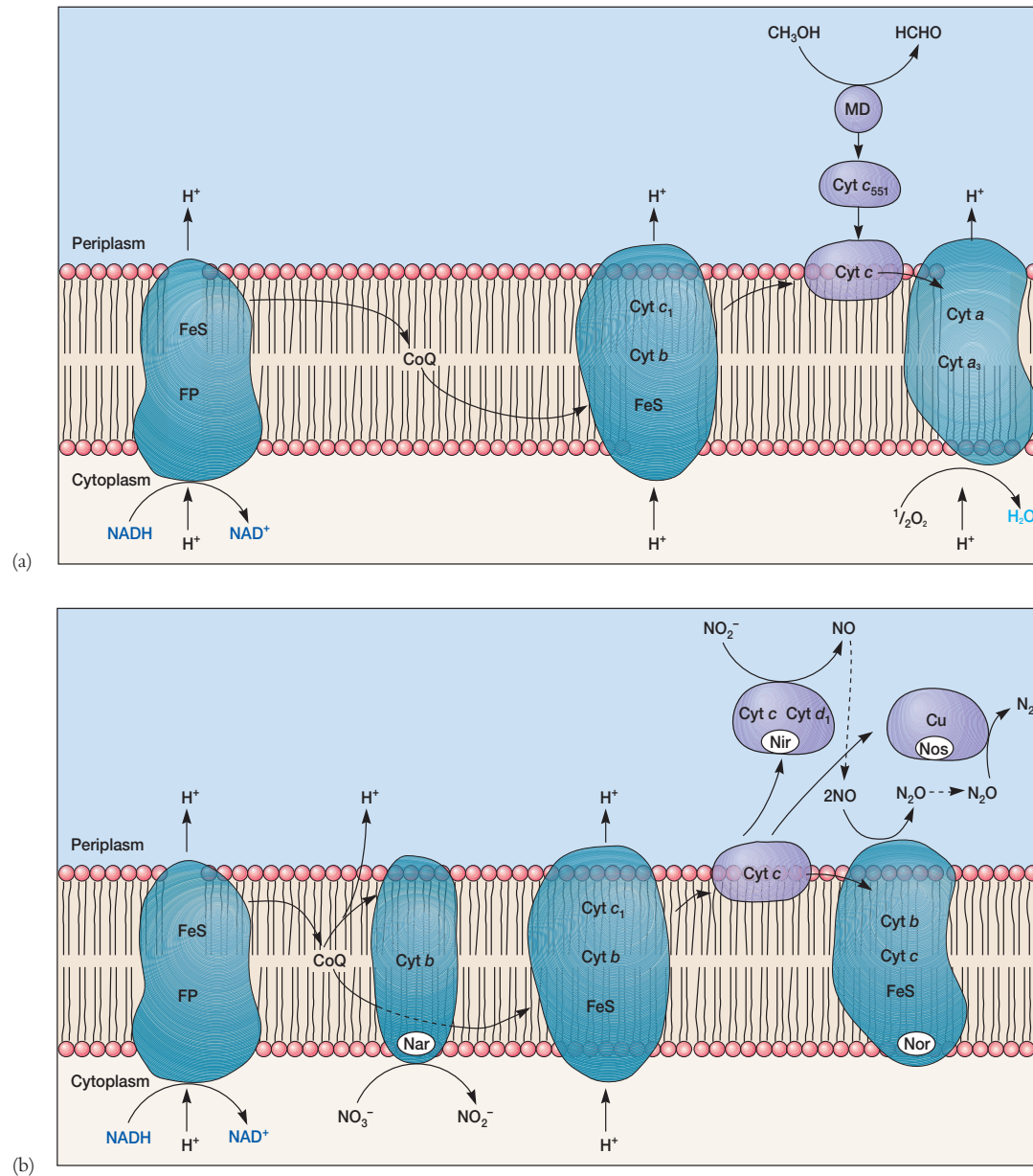


Figure 9.16 *Paracoccus denitrificans* Electron Transport Chains. (a) The aerobic transport chain resembles a mitochondrial electron transport chain and uses oxygen as its acceptor. Methanol and methylamine can contribute electrons at the cytochrome *c* level. (b) The highly branched anaerobic chain is made of both membrane and periplasmic proteins. Nitrate is reduced to diatomic nitrogen by the collective action of four different reductases that receive electrons from CoQ and cytochrome *c*. Locations of proton movement are shown, but the number of protons involved has not been indicated. Abbreviations used: flavoprotein (FP), methanol dehydrogenase (MD), nitrate reductase (Nar), nitrite reductase (Nir), nitric oxide reductase (Nor), and nitrous oxide reductase (Nos).

Paracoccus denitrificans is a gram-negative, facultative anaerobic soil bacterium that can grow heterotrophically with a variety of nutrients or autotrophically on H₂ and CO₂ with NO₃⁻ as the electron acceptor. The bacterium carries out either aerobic respiration or anaerobic respiration with nitrate as an acceptor. The aero-

bic electron transport chain has four complexes that correspond to the mitochondrial chain (**figure 9.16a**). In addition to donors such as NADH and succinate, *Paracoccus* oxidizes methanol and methylamine and grows with them as the sole carbon source. The electrons enter the transport chain at the cytochrome *c* level.

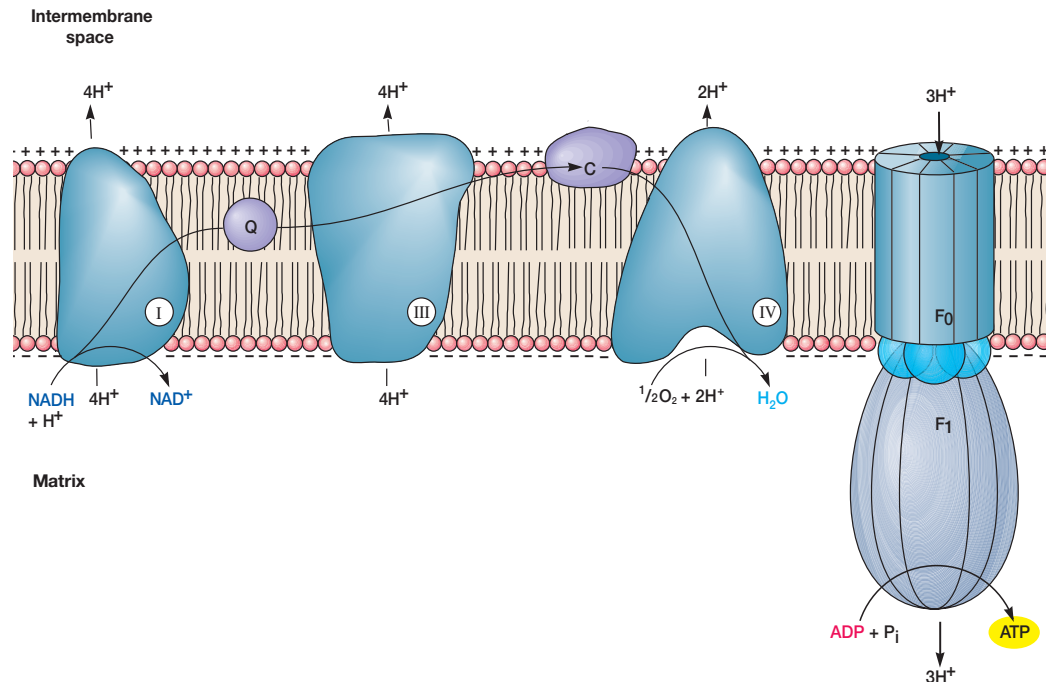


Figure 9.17 Chemiosmosis. An overview of the chemiosmotic hypothesis as applied to mitochondrial function. The flow of electrons from NADH to oxygen causes protons to move from the mitochondrial matrix to the intermembrane space. This generates proton and electrical gradients. When protons move back to the matrix through the F_1F_0 complex, F_1 synthesizes ATP. In procaryotes, the process is similar except that the protons move from the cytoplasm to the periplasm.

Methanol is oxidized to formaldehyde, which is converted to CO_2 and incorporated by the Calvin cycle (*see pp. 207–8*). When the bacterium is growing anaerobically with nitrate as its electron acceptor, the chain is structured quite differently (figure 9.16b). The cytochrome aa_3 complex does not function. Rather electrons from the cytochrome c level of the chain move to nitrite reductase, nitric oxide reductase, and nitrous oxide reductase. Nitrate reductase is supplied electrons by coenzyme Q. Not as many protons cross the membrane with this arrangement, but it does allow anaerobic growth. We will return to this process in the context of anaerobic respiration and denitrification. [Anaerobic respiration \(pp. 190–91\)](#)

Oxidative Phosphorylation

The mechanism by which oxidative phosphorylation takes place has been studied intensively for years. Currently the most widely accepted hypothesis about how oxidative phosphorylation occurs is the chemiosmotic hypothesis. For the sake of clarity, we will confine our attention to this hypothesis.

According to the **chemiosmotic hypothesis**, first formulated in 1961 by the British biochemist Peter Mitchell, the electron transport chain is organized so that protons move outward from the mitochondrial matrix and electrons are transported inward (figure 9.14; **figure 9.17**). Proton movement may result ei-

ther from carrier loops, as shown in figure 9.14, or from the action of special proton pumps that derive their energy from electron transport. The result is **proton motive force (PMF)**, composed of a gradient of protons and a membrane potential due to the unequal distribution of charges. When protons return to the mitochondrial matrix driven by the proton motive force, ATP is synthesized in a reversal of the ATP hydrolysis reaction (figure 9.17). A similar process takes place in procaryotes, with electron flow causing the protons to move outward across the plasma membrane (figures 9.15 and 9.16). ATP synthesis occurs when these protons diffuse back into the cell. The proton motive force also may drive the transport of molecules across membranes (*see section 5.6*) and the rotation of bacterial flagella (*see section 3.6*) and thus plays a central role in procaryotic physiology (**figure 9.18**). The chemiosmotic hypothesis is accepted by most microbiologists. There is considerable evidence for the generation of proton and charge gradients across membranes. However, the evidence for proton gradients as the direct driving force for oxidative phosphorylation is not yet conclusive. In some halophilic marine bacteria, sodium ions may be used to drive ATP synthesis.

Whatever the precise mechanism, ATP synthesis takes place at the F_1F_0 ATPase or **ATP synthase** (**figure 9.19**). The mitochondrial F_1 component appears as a spherical structure attached

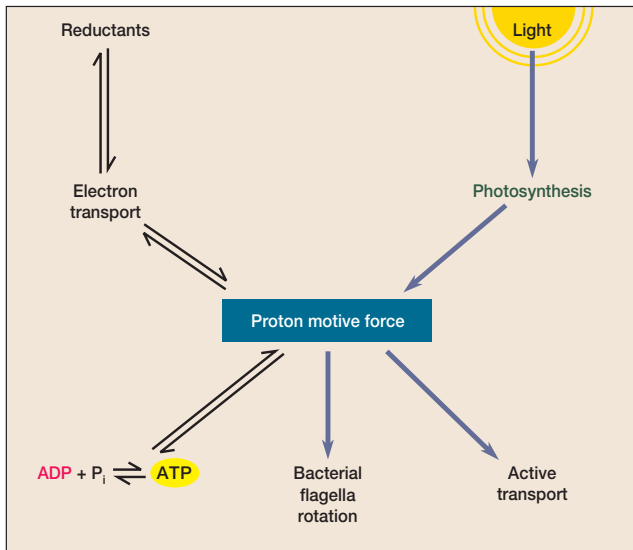
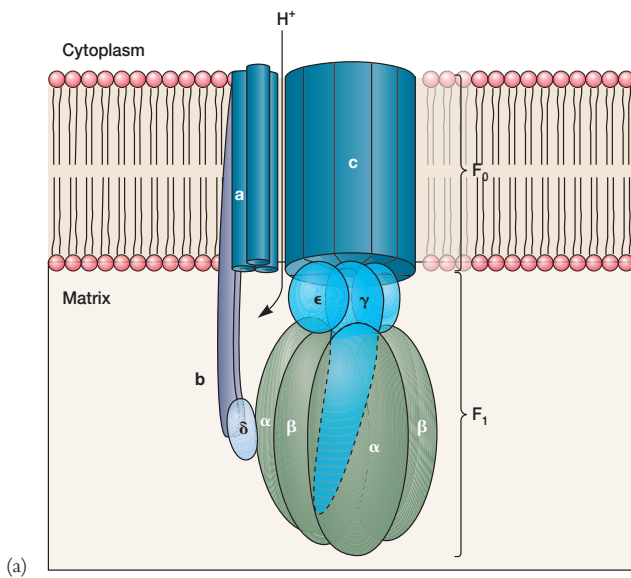
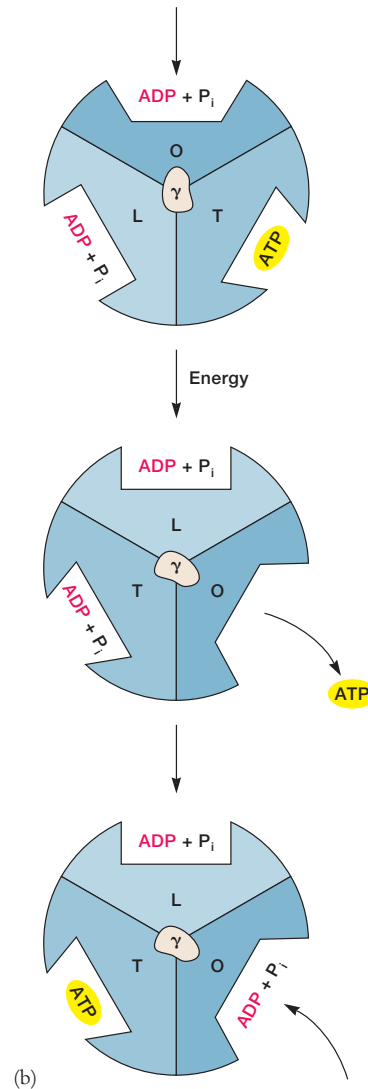


Figure 9.18 The Central Role of Proton Motive Force. It should be noted that active transport is not always driven by PMF.



(a)



(b)

Figure 9.19 ATP Synthase Structure and Function. (a) The major structural features of ATP synthase deduced from X-ray crystallography and other studies. F₁ is a spherical structure composed largely of alternating α and β subunits; the three active sites are on the β subunits. The γ subunit extends upward through the center of the sphere and can rotate. The stalk (γ and ε subunits) connects the sphere to F₀, the membrane embedded complex that serves as a proton channel. F₀ contains one a subunit, two b subunits, and 9–12 c subunits. The stator arm is composed of subunit a, two b subunits, and the δ subunit; it is embedded in the membrane and attached to F₁. A ring of c subunits in F₀ is connected to the stalk and may act as a rotor and move past the a subunit of the stator. As the c subunit ring turns, it rotates the shaft (γ and ε subunits). (b) The binding change mechanism of ATP synthesis is depicted in a simplified schematic diagram in which the F₁ sphere is viewed from the membrane side. The three active sites appear able to exist in three different conformations: an inactive open (O) conformation with low affinity for substrates, an inactive L conformation with fairly loose affinity for the substrates, and an active tight (T) conformation that has high affinity for substrates. In the first step, ADP and P_i bind to the O site. Then the γ subunit rotates 120° with the input of energy, presumably from proton flow through F₀. This rotation causes conformational changes in all three subunits resulting in a release of newly formed ATP and a conversion of the L site to an active T conformation. Finally, ATP is formed at the new T site while more ADP and P_i bind to the unoccupied O site and everything is ready for another energy-driven γ subunit rotation.

to the inner membrane surface by a stalk and the F_0 component, which is embedded in the membrane. The F_1F_0 ATPase is on the inner surface of the plasma membrane in bacteria. F_0 participates in proton movement across the membrane, and this movement through a channel in F_0 is believed to drive oxidative phosphorylation. F_1 is a large complex in which three α subunits alternate with three β subunits. The γ subunit extends downward from the $\alpha_3\beta_3$ complex; it composes part of the stalk and interacts with F_0 . The δ subunit also is located in the stalk. Much of the γ subunit is positioned in the center of F_1 , surrounded by the α and β subunits. The γ subunit rotates rapidly in a counterclockwise direction within the $\alpha_3\beta_3$ complex much like a car's crankshaft and causes conformational changes that drive ATP synthesis at the active sites on the β subunits (figure 9.19b). Thus the ATP synthase is the smallest rotary motor known, much smaller than the bacterial flagellum.

Many chemicals inhibit the aerobic synthesis of ATP and can even kill cells at sufficiently high concentrations. These inhibitors generally fall into two categories. Some directly block the transport of electrons. The antibiotic piericidin competes with coenzyme Q; the antibiotic antimycin A blocks electron transport between cytochromes *b* and *c*; and both cyanide and azide stop the transfer of electrons between cytochrome *a* and O_2 because they are structural analogs of O_2 . Another group of inhibitors known as **uncouplers** stops ATP synthesis without inhibiting electron transport itself. Indeed, they may even enhance the rate of electron flow. Normally electron transport is tightly coupled with oxidative phosphorylation so that the rate of ATP synthesis controls the rate of electron transport. The more rapidly ATP is synthesized during oxidative phosphorylation, the faster the electron transport chain operates to supply the required energy. Uncouplers disconnect oxidative phosphorylation from electron transport; therefore the energy released by the chain is given off as heat rather than as ATP. Many uncouplers like dinitrophenol and valinomycin may allow hydrogen ions, potassium ions, and other ions to cross the membrane without activating the F_1F_0 ATPase. In this way they destroy the pH and ion gradients. Valinomycin also may bind directly to the F_1F_0 ATPase and inhibit its activity.

The Yield of ATP in Glycolysis and Aerobic Respiration

The maximum ATP yield in eucaryotes from glycolysis, the TCA cycle, and electron transport can be readily calculated. The conversion of glucose to two pyruvate molecules during glycolysis gives a net gain of two ATPs and two NADHs. Because each NADH can yield a maximum of three ATPs during electron transport and oxidative phosphorylation (a P/O ratio of 3), the total aerobic yield from the glycolytic pathway is eight ATP molecules (**table 9.2**). Under anaerobic conditions, when the NADH is not oxidized by the electron transport chain, only two ATPs will be generated during the degradation of glucose to pyruvate.

When O_2 is present and the electron transport chain is operating, pyruvate is next oxidized to acetyl-CoA, the substrate for the TCA cycle. This reaction yields 2 NADHs because 2 pyruvates arise from a glucose; therefore 6 more ATPs are

Table 9.2 ATP Yield from the Aerobic Oxidation of Glucose by Eucaryotic Cells

Glycolytic Pathway	
Substrate-level phosphorylation (ATP)	2 ATP ^a
Oxidative phosphorylation with 2 NADH	6 ATP
2 Pyruvate to 2 Acetyl-CoA	
Oxidative phosphorylation with 2 NADH	6 ATP
Tricarboxylic Acid Cycle	
Substrate-level phosphorylation (GTP)	2 ATP
Oxidative phosphorylation with 6 NADH	18 ATP
Oxidative phosphorylation with 2 FADH ₂	4 ATP
Total Aerobic Yield	38 ATP

^aATP yields are calculated with an assumed P/O ratio of 3.0 for NADH and 2.0 for FADH₂.

formed. Oxidation of each acetyl-CoA in the TCA cycle will yield 1 GTP (or ATP), 3 NADHs, and a single FADH₂ for a total of 2 GTPs (ATPs), 6 NADHs, and 2 FADH₂s from two acetyl-CoA molecules. As table 9.2 shows, this amounts to 24 ATPs when NADH and FADH₂ from the cycle are oxidized in the electron transport chain. Thus the aerobic oxidation of glucose to 6 CO_2 molecules supplies a maximum of 38 ATPs. The calculations just summarized and presented in table 9.2 are theoretical and based on P/O ratios (the number of ATPs formed per oxygen atom reduced by 2 electrons in electron transport) of 3.0 for NADH oxidation and 2.0 for FADH₂. In fact, the P/O ratios are more likely about 2.5 for NADH and 1.5 for FADH₂. Thus the total ATP aerobic yield from glucose may be closer to 30 ATPs rather than 38.

Because bacterial electron transport systems often have lower P/O ratios than the eucaryotic system being discussed, bacterial aerobic ATP yields can be less. For example, *E. coli* with its truncated electron transport chains has a P/O ratio around 1.3 when using the cytochrome *bo* path at high oxygen levels and only a ratio of about 0.67 when employing the cytochrome *bd* branch (figure 9.15) at low oxygen concentrations. In this case ATP production varies with environmental conditions. Perhaps because *E. coli* normally grows in habitats such as the intestinal tract that are very rich in nutrients, it does not have to be particularly efficient in ATP synthesis. Presumably the transport chain functions when *E. coli* is in an aerobic freshwater environment between hosts.

Clearly, aerobic respiration is much more effective than anaerobic processes not involving electron transport and oxidative phosphorylation. Many microorganisms, when moved from anaerobic to aerobic conditions, will drastically reduce their rate of sugar catabolism and switch to aerobic respiration, a regulatory phenomenon known as the **Pasteur effect**. This is of obvious advantage to the microorganism as less sugar must be degraded to obtain the same amount of ATP when the more efficient aerobic process can be employed.

Table 9.3 Some Electron Acceptors
Used in Respiration

	Electron Acceptor	Reduced Products	Examples of Microorganisms
Aerobic	O ₂	H ₂ O	All aerobic bacteria, fungi, protozoa, and algae
Anaerobic	NO ₃ ⁻	NO ₂ ⁻	Enteric bacteria
	NO ₃ ⁻	NO ₂ ⁻ , N ₂ O, N ₂	<i>Pseudomonas</i> , <i>Bacillus</i> , and <i>Paracoccus</i>
	SO ₄ ²⁻	H ₂ S	<i>Desulfovibrio</i> and <i>Desulfotomaculum</i>
	CO ₂	CH ₄	All methanogens
	S ⁰	H ₂ S	<i>Desulfuromonas</i> and <i>Thermoproteus</i>
	Fe ³⁺	Fe ²⁺	<i>Pseudomonas</i> , <i>Bacillus</i> , and <i>Geobacter</i>
	HAsO ₄ ²⁻	HAsO ₂	<i>Bacillus</i> , <i>Desulfotomaculum</i> , <i>Sulfurospirillum</i>
	SeO ₄ ²⁻	Se, HSeO ₃ ⁻	<i>Aeromonas</i> , <i>Bacillus</i> , <i>Thauera</i>
	Fumarate	Succinate	<i>Wolinella</i>

1. Briefly describe the structure of the electron transport chain and its role in ATP formation. How do mitochondrial and bacterial chains differ?
2. By what mechanism might ATP be synthesized during oxidative phosphorylation? Briefly describe the structure of ATP synthase and how it functions. What is an uncoupler?
3. Calculate the ATP yield in eucaryotes for both glycolysis and the total aerobic oxidation of glucose. Explain your reasoning.

9.6 Anaerobic Respiration

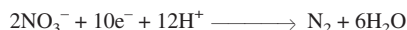
Electrons derived from sugars and other organic molecules are usually donated either to endogenous organic electron acceptors or to molecular O₂ by way of an electron transport chain (figure 9.2). However, many bacteria have electron transport chains that can operate with exogenous electron acceptors other than O₂. As noted earlier, this energy-yielding process is called anaerobic respiration. The major electron acceptors are nitrate, sulfate, and CO₂, but metals and a few organic molecules can also be reduced (**table 9.3**).

Some bacteria can use nitrate as the electron acceptor at the end of their electron transport chain and still produce ATP. Often this process is called **dissimilatory nitrate reduction**. Nitrate may be reduced to nitrite by nitrate reductase, which replaces cytochrome oxidase.

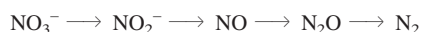


However, reduction of nitrate to nitrite is not a particularly effective way of making ATP, because a large amount of nitrate is required for growth (a nitrate molecule will accept only two electrons). The nitrite formed is also quite toxic. Therefore nitrate

often is further reduced all the way to nitrogen gas, a process known as **denitrification**. Each nitrate will then accept five electrons, and the product will be nontoxic.



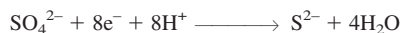
There is considerable evidence that denitrification is a multistep process with four enzymes participating: nitrate reductase, nitrite reductase, nitric oxide reductase, and nitrous oxide reductase.



Interestingly, one of the intermediates is nitric oxide (NO). In mammals this molecule acts as a neurotransmitter, helps regulate blood pressure, and is used by macrophages to destroy bacteria and tumor cells (*see p. 720*). Two types of bacterial nitrite reductases catalyze the formation of NO in bacteria. One contains cytochromes *c* and *d*₁ (e.g., *Paracoccus* and *Pseudomonas aeruginosa*), and the other is a copper protein (e.g., *Alcaligenes*). Nitrite reductase seems to be periplasmic in gram-negative bacteria. Nitric oxide reductase catalyzes the formation of nitrous oxide from NO and is a membrane-bound cytochrome *bc* complex. A well-studied example of denitrification is gram-negative soil bacterium *Paracoccus denitrificans*, which reduces nitrate to N₂ anaerobically. Its chain contains membrane-bound nitrate reductase and nitric oxide reductase, whereas nitrite reductase and nitrous oxide reductase are periplasmic (figure 9.16*b*). The four enzymes use electrons from coenzyme Q and *c*-type cytochromes to reduce nitrate and generate PMF. *Paracoccus denitrificans* electron transport (p. 186)

Denitrification is carried out by some members of the genera *Pseudomonas*, *Paracoccus*, and *Bacillus*. They use this route as an alternative to normal aerobic respiration and may be considered facultative anaerobes. If O₂ is present, these bacteria use aerobic respiration (the synthesis of nitrate reductase is repressed by O₂). Denitrification in anaerobic soil results in the loss of soil nitrogen and adversely affects soil fertility. *The nitrogen cycle and denitrification* (pp. 615–16)

Two other major groups of bacteria employing anaerobic respiration are obligate anaerobes. Those using CO₂ or carbonate as a terminal electron acceptor are called methanogens because they reduce CO₂ to methane (*see chapter 20*). Sulfate also can act as the final acceptor in bacteria such as *Desulfovibrio*. It is reduced to sulfide (S²⁻ or H₂S), and eight electrons are accepted.



Anaerobic respiration is not as efficient in ATP synthesis as aerobic respiration—that is, not as much ATP is produced by oxidative phosphorylation with nitrate, sulfate, or CO₂ as the terminal acceptors. Reduction in ATP yield arises from the fact that these alternate electron acceptors have less positive reduction potentials than O₂ (*see table 8.1*). The reduction potential difference between a donor like NADH and nitrate is smaller than the difference between NADH and O₂. Because energy yield is directly related to the magnitude of the reduction potential difference, less energy is available to make ATP in anaerobic respiration. Nevertheless, anaerobic respiration is useful because it is more efficient than fermentation and allows ATP synthesis by electron transport and ox-

idative phosphorylation in the absence of O_2 . Anaerobic respiration is very prevalent in oxygen-depleted soils and sediments.

Often one will see a succession of microorganisms in an environment when several electron acceptors are present. For example, if O_2 , nitrate, manganese ion, ferric ion, sulfate, and CO_2 are available in a particular environment, a predictable sequence of oxidant use takes place when an oxidizable substrate is available to the microbial population. Oxygen is employed as an electron acceptor first because it inhibits nitrate use by microorganisms capable of respiration with either O_2 or nitrate. While O_2 is available, sulfate reducers and methanogens are inhibited because these groups are obligate anaerobes.

Once the O_2 and nitrate are exhausted and fermentation products, including hydrogen, have accumulated, competition for use of other oxidants begins. Manganese and iron will be used first, followed by competition between sulfate reducers and methanogens. This competition is influenced by the greater energy yield obtained with sulfate as an electron acceptor. Differences in enzymatic affinity for hydrogen, an important substrate used by both groups (section 9.10), also are important. The sulfate reducer *Desulfovibrio* grows rapidly and uses the available hydrogen at a faster rate than *Methanobacterium*. When the sulfate is exhausted, *Desulfovibrio* no longer oxidizes hydrogen, and the hydrogen concentration rises. The methanogens finally dominate the habitat and reduce CO_2 to methane.

1. Describe the process of anaerobic respiration. Is as much ATP produced in anaerobic respiration as in aerobic respiration? Why or why not?
2. What is denitrification?

9.7 Catabolism of Carbohydrates and Intracellular Reserve Polymers

Microorganisms can catabolize many carbohydrates besides glucose. These carbohydrates may come either from outside the cell or from internal sources. Often the initial steps in the degradation of external carbohydrate polymers differ from those employed with internal reserves.

Carbohydrates

Figure 9.20 outlines some catabolic pathways for the monosaccharides (single sugars) glucose, fructose, mannose, and galactose. The first three are phosphorylated using ATP and easily enter the glycolytic pathway. In contrast, galactose must be converted to uridine diphosphate galactose (*see p. 209*) after initial phosphorylation, then changed into glucose 6-phosphate in a three-step process (figure 9.20).

The common disaccharides are cleaved to monosaccharides by at least two mechanisms (figure 9.20). Maltose, sucrose, and lactose can be directly hydrolyzed to their constituent sugars. Many disaccharides (e.g., maltose, cellobiose, and sucrose) are

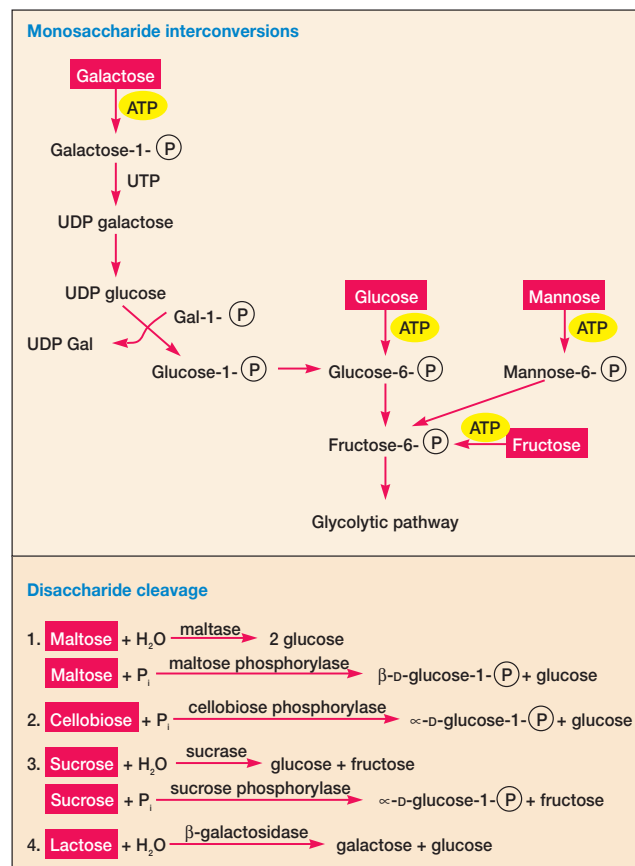


Figure 9.20 Carbohydrate Catabolism. Examples of enzymes and pathways used in disaccharide and monosaccharide catabolism. UDP is an abbreviation for uridine diphosphate.

also split by a phosphate attack on the bond joining the two sugars, a process called phosphorolysis.

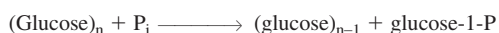
Polysaccharides, like disaccharides, are cleaved by both hydrolysis and phosphorolysis. Bacteria and fungi degrade external polysaccharides by secreting hydrolytic enzymes that cleave polysaccharides into smaller molecules, which can then be assimilated. Starch and glycogen are hydrolyzed by amylases to glucose, maltose, and other products. Cellulose is more difficult to digest; many fungi and a few bacteria (some gliding bacteria, clostridia, and actinomycetes) produce cellulases that hydrolyze cellulose to cellobiose and glucose. Some members of the bacterial genus *Cytophaga*, isolated from marine habitats, excrete an agarase that degrades agar. Many soil bacteria and bacterial plant pathogens degrade pectin, a polymer of galacturonic acid (a galactose derivative) that is an important constituent of plant cell walls and tissues.

In the context of compounds that are recalcitrant or difficult to digest, it should be noted that microorganisms also can degrade xenobiotic compounds (foreign substances not formed by natural biosynthetic processes) such as pesticides and various aromatic

compounds. They transform these molecules to normal metabolic intermediates by use of special enzymes and pathways, then continue catabolism in the usual way. Biodegradation and bioremediation are discussed in chapter 42; the fungus *Phanerochaete chrysosporium* is an extraordinary example of the ability to degrade xenobiotics (see Box 42.3).

Reserve Polymers

Microorganisms often survive for long periods in the absence of exogenous nutrients. Under such circumstances they catabolize intracellular stores of glycogen, starch, poly- β -hydroxybutyrate, and other energy reserves. Glycogen and starch are degraded by phosphorylases. Phosphorylases catalyze a phosphorylysis reaction that shortens the polysaccharide chain by one glucose and yields glucose 1-phosphate.



Glucose 1-phosphate can enter the glycolytic pathway by way of glucose 6-phosphate (figure 9.20).

Poly- β -hydroxybutyrate (PHB) is an important, wide-spread reserve material. Its catabolism has been studied most thoroughly in *Azotobacter*. This bacterium hydrolyzes PHB to 3-hydroxybutyrate, then oxidizes the hydroxybutyrate to acetoacetate. Acetoacetate is converted to acetyl-CoA, which can be oxidized in the TCA cycle.

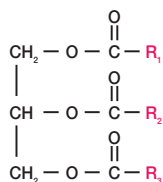


Figure 9.21 A Triacylglycerol or Triglyceride. The R groups represent the fatty acid side chains.

9.8 Lipid Catabolism

Microorganisms frequently use lipids as energy sources. Triglycerides or triacylglycerols, esters of glycerol and fatty acids (**figure 9.21**), are common energy sources and will serve as our examples. They can be hydrolyzed to glycerol and fatty acids by microbial lipases. The glycerol is then phosphorylated, oxidized to dihydroxyacetone phosphate, and catabolized in the glycolytic pathway (figure 9.5).

Fatty acids from triacylglycerols and other lipids are often oxidized in the **β -oxidation pathway** after conversion to coenzyme A esters (**figure 9.22**). In this cyclic pathway fatty acids are degraded to acetyl-CoA, which can be fed into the TCA cycle or used in biosynthesis (see section 10.8). One turn of the cycle produces acetyl-CoA, NADH, and FADH₂; NADH and FADH₂ can be oxidized by the electron transport chain to provide more ATP. The fatty acyl-CoA, shortened by two carbons, is ready for another turn of the cycle. Lipid fatty acids are a rich source of energy for microbial growth. In a similar fashion some microorganisms grow well on petroleum hydrocarbons under aerobic conditions.

9.9 Protein and Amino Acid Catabolism

Some bacteria and fungi—particularly pathogenic, food spoilage, and soil microorganisms—can use proteins as their source of carbon and energy. They secrete **protease** enzymes that hydrolyze proteins and polypeptides to amino acids, which are transported into the cell and catabolized.

The first step in amino acid use is **deamination**, the removal of the amino group from an amino acid. This is often accomplished by **transamination**. The amino group is transferred from an amino acid to an α -keto acid acceptor (**figure 9.23**). The organic acid resulting from deamination can be converted to pyruvate, acetyl-CoA, or a TCA cycle intermediate and eventually oxidized in the TCA cycle to release energy. It also can be used as a

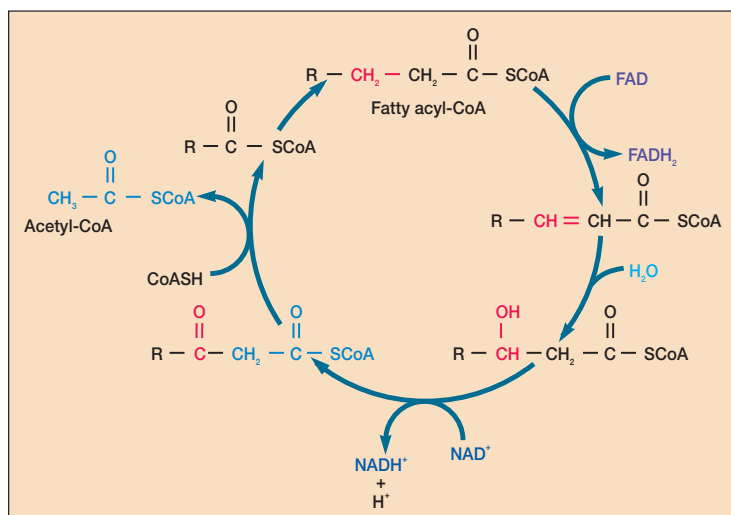


Figure 9.22 Fatty Acid β -Oxidation. The portions of the fatty acid being modified are shown in color.

source of carbon for the synthesis of cell constituents. Excess nitrogen from deamination may be excreted as ammonium ion, thus making the medium alkaline.

1. Briefly discuss the ways in which microorganisms degrade and use common monosaccharides, disaccharides, and polysaccharides from both external and internal sources.
2. Describe how a microorganism might derive carbon and energy from the lipids and proteins in its diet. What is β -oxidation? Transamination?

9.10 Oxidation of Inorganic Molecules

As we have seen, microorganisms can oxidize organic molecules such as carbohydrates, lipids, and proteins and synthesize ATP with the energy liberated. The electron acceptor is (1) another more oxidized endogenous organic molecule in fermentation, (2) O_2 in aerobic respiration, or (3) an oxidized exogenous molecule other than O_2 in anaerobic respiration (figure 9.2). In both aerobic and anaerobic respiration, ATP is formed as a result of electron transport chain activity. Electrons for the chain can be obtained from inorganic nutrients, and it is possible to derive energy from the oxidation of inorganic molecules rather than from organic nutrients. This ability is confined to a group of bacteria called **chemolithotrophs** (see sections 22.1 and 22.2). Each species is rather specific in its preferences for electron donors and

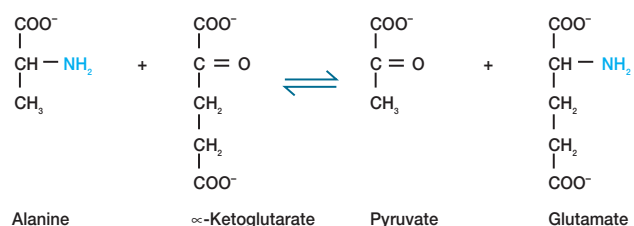
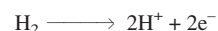


Figure 9.23 Transamination. A common example of this process. The α -amino group (blue) of alanine is transferred to the acceptor α -ketoglutarate forming pyruvate and glutamate. The pyruvate can be catabolized in the tricarboxylic acid cycle or used in biosynthesis.

acceptors (table 9.4). The acceptor is usually O_2 , but sulfate and nitrate are also used. The most common electron donors are hydrogen, reduced nitrogen compounds, reduced sulfur compounds, and ferrous iron (Fe^{2+}).

Chemolithotrophic bacteria are usually autotrophic and employ the Calvin cycle to fix CO_2 as their carbon source (see section 10.2). However, some chemolithotrophs can function as heterotrophs if reduced organic compounds are available. Considerable energy is required to reduce CO_2 to carbohydrate; incorporation of one CO_2 in the Calvin cycle requires three ATPs and two NADPHs. Moreover, much less energy is available from the oxidation of inorganic molecules (table 9.5) than from the complete oxidation of glucose to CO_2 , which is accompanied by a standard free energy change of -686 kcal/mole. The P/O ratios for oxidative phosphorylation in chemolithotrophs are probably around 1.0 (in the oxidation of hydrogen it is significantly higher). Because the yield of ATP is so low, chemolithotrophs must oxidize a large quantity of inorganic material to grow and reproduce, and this magnifies their ecological impact.

Several bacterial genera (table 9.4) can oxidize hydrogen gas to produce energy because they possess a hydrogenase enzyme that catalyzes the oxidation of hydrogen.

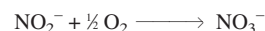


The electrons are donated either to an electron transport chain or to NAD^+ , depending on the hydrogenase. If NADH is produced, it can be used in ATP synthesis by electron transport and oxidative phosphorylation, with O_2 as the terminal electron acceptor. These hydrogen-oxidizing microorganisms often will use organic compounds as energy sources when such nutrients are available.

The best-studied nitrogen-oxidizing chemolithotrophs are the **nitrifying bacteria** (see section 22.1). These are soil and aquatic bacteria of considerable ecological significance. Ammonia oxidation to nitrate depends on the activity of at least two different genera. For example, *Nitrosomonas* and *Nitrosospira* oxidize ammonia to nitrite.



The nitrite can then be further oxidized by *Nitrobacter* and *Nitrococcus* to yield nitrate.



When two genera work together, ammonia in the soil is oxidized to nitrate in a process called **nitrification**. The role of chemolithotrophs in soil and aquatic ecosystems (pp. 611–18)

Table 9.4 Representative Chemolithotrophs and Their Energy Sources

Bacteria	Electron Donor	Electron Acceptor	Products
<i>Alcaligenes</i> , <i>Hydrogenophaga</i> , and <i>Pseudomonas</i> spp.	H_2	O_2	H_2O
<i>Nitrobacter</i>	NO_2^-	O_2	NO_3^- , H_2O
<i>Nitrosomonas</i>	NH_4^+	O_2	NO_2^- , H_2O
<i>Thiobacillus denitrificans</i>	S^0 , H_2S	NO_3^-	SO_4^{2-} , N_2
<i>Thiobacillus ferrooxidans</i>	Fe^{2+} , S^0 , H_2S	O_2	Fe^{3+} , H_2O , H_2SO_4

Energy released upon the oxidation of both ammonia and nitrite is used to make ATP by oxidative phosphorylation. However, microorganisms need a source of electrons (reducing power) as well as a source of ATP in order to reduce CO_2 and other molecules. Since molecules like ammonia and nitrite have more positive reduction potentials than NAD^+ , they cannot directly donate their electrons to form the required NADH and NADPH. This is because electrons spontaneously move only from donors with more negative reduction potentials to acceptors with more positive potentials (see section 8.5 and figure 8.7). Sulfur-oxidizing bacteria face the same difficulty. Both types of chemolithotrophs solve this problem by using proton motive force to reverse the flow of electrons in their electron transport chains and reduce NAD^+ with

electrons from nitrogen and sulfur donors (figure 9.24). Because energy is used to generate NADH as well as ATP, the net yield of ATP is fairly low. Chemolithotrophs can afford this inefficiency as they have no serious competitors for their unique energy sources.

The sulfur-oxidizing bacteria are the third major group of chemolithotrophs. The metabolism of *Thiobacillus* has been best studied. These bacteria oxidize sulfur (S^0), hydrogen sulfide (H_2S), thiosulfate ($\text{S}_2\text{O}_3^{2-}$), and other reduced sulfur compounds to sulfuric acid; therefore they have a significant ecological impact (Box 9.2). Interestingly they generate ATP by both oxidative phosphorylation and substrate-level phosphorylation involving **adenosine 5'-phosphosulfate (APS)**. APS is a high-energy molecule formed from sulfite and adenosine monophosphate (figure 9.25).

Some of these procaryotes are extraordinarily flexible metabolically. For example, *Sulfolobus brierleyi* and a few other species can grow aerobically as sulfur-oxidizing bacteria; in the absence of O_2 , they carry out anaerobic respiration with molecular sulfur as an electron acceptor.

Sulfur-oxidizing bacteria, like other chemolithotrophs, can use CO_2 as their carbon source. Many will grow heterotrophically if they are supplied with reduced organic carbon sources like glucose or amino acids.

Table 9.5 Energy Yields from Oxidations Used by Chemolithotrophs

Reaction	$\Delta G^{\circ\prime}$ (kcal/mole) ^a
$\text{H}_2 + \frac{1}{2} \text{O}_2 \longrightarrow \text{H}_2\text{O}$	-56.6
$\text{NO}_2^- + \frac{1}{2} \text{O}_2 \longrightarrow \text{NO}_3^-$	-17.4
$\text{NH}_4^+ + 1\frac{1}{2} \text{O}_2 \longrightarrow \text{NO}_2^- + \text{H}_2\text{O} + 2\text{H}^+$	-65.0
$\text{S}^0 + 1\frac{1}{2} \text{O}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4$	-118.5
$\text{S}_2\text{O}_3^{2-} + 2\text{O}_2 + \text{H}_2\text{O} \longrightarrow 2\text{SO}_4^{2-} + 2\text{H}^+$	-223.7
$2\text{Fe}^{2+} + 2\text{H}^+ + \frac{1}{2} \text{O}_2 \longrightarrow 2\text{Fe}^{3+} + \text{H}_2\text{O}$	-11.2

^aThe $\Delta G^{\circ\prime}$ for complete oxidation of glucose to CO_2 is -686 kcal/mole. A kcal is equivalent to 4.184 kJ.

1. How do chemolithotrophs obtain their ATP and NADH? What is their source of carbon?
2. Describe energy production by hydrogen-oxidizing bacteria, nitrifying bacteria, and sulfur-oxidizing bacteria.

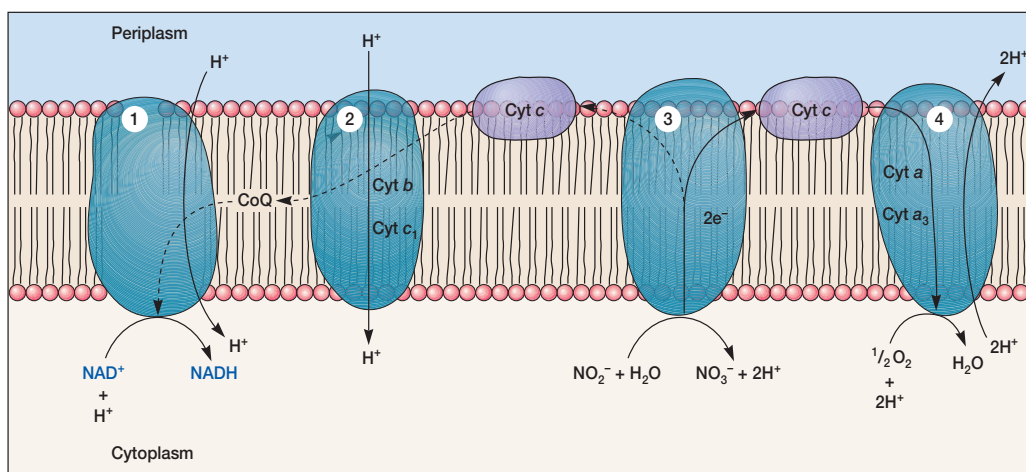


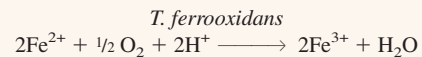
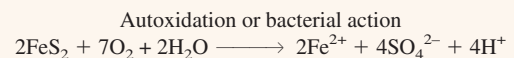
Figure 9.24 Electron Flow in *Nitrobacter* Electron Transport Chain. *Nitrobacter* carries out normal electron transport to generate proton motive force for ATP synthesis. This is the right-hand branch of the diagram. Some of the proton motive force also is used to force electrons to flow up the reduction potential gradient from nitrite to NAD^+ (left-hand branch). Cytochrome *c* and four complexes are involved: NADH-ubiquinone oxidoreductase (1), ubiquinol-cytochrome *c* oxidoreductase (2), nitrite oxidase (3), and cytochrome *aa*₃ oxidase (4).

Box 9.2

Acid Mine Drainage

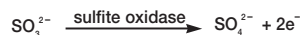
Each year millions of tons of sulfuric acid flow down the Ohio River from the Appalachian Mountains. This sulfuric acid is of microbial origin and leaches enough metals from the mines to make the river reddish and acidic. The primary culprit is *Thiobacillus ferrooxidans*, a chemolithotrophic bacterium that derives its energy from oxidizing ferrous ion to ferric ion and sulfide ion to sulfate ion. The combination of these two energy sources is important because of the solubility properties of iron. Ferrous ion is somewhat soluble and can be formed at pH values of 3.0 or less in moderately reducing environments. However, when the pH is greater than 4.0 to 5.0, ferrous ion is spontaneously oxidized to ferric ion by O_2 in the water and precipitates as a hydroxide. If the pH drops below about 2.0 to 3.0 because of sulfuric acid production by spontaneous oxidation of sulfur or sulfur oxidation by thiobacilli and other bacteria, the ferrous ion remains reduced, soluble, and available as an energy source. Remarkably, *T. ferrooxidans* grows well at such acid pHs and actively oxidizes ferrous ion to an insoluble ferric precipitate. The water is rendered toxic for most aquatic life and unfit for human consumption.

The ecological consequences of this metabolic life-style arise from the common presence of pyrite (FeS_2) in coal mines. The bacteria oxidize both elemental components of pyrite for their growth and in the process form sulfuric acid, which leaches the remaining minerals.

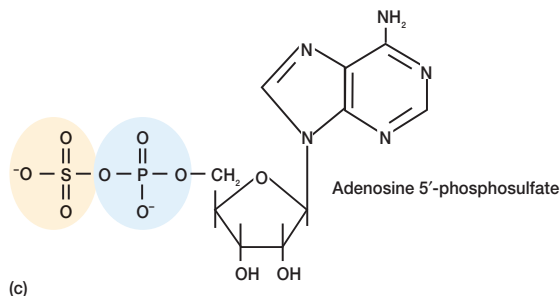
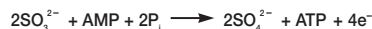
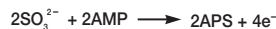


Pyrite oxidation is further accelerated because the ferric ion generated by bacterial activity readily oxidizes more pyrite to sulfuric acid and ferrous ion. In turn the ferrous ion supports further bacterial growth. It is difficult to prevent *T. ferrooxidans* growth as it requires only pyrite and common inorganic salts. Because *T. ferrooxidans* gets its O_2 and CO_2 from the air, the only feasible method of preventing its damaging growth is to seal the mines to render the habitat anaerobic.

(a) Direct oxidation of sulfite



(b) Formation of adenosine 5'-phosphosulfate



(c)

Figure 9.25 Energy Generation by Sulfur Oxidation. (a) Sulfite can be directly oxidized to provide electrons for electron transport and oxidative phosphorylation. (b) Sulfite can also be oxidized and converted to APS. This route produces electrons for use in electron transport and ATP by substrate-level phosphorylation with APS. (c) The structure of adenosine 5'-phosphosulfate.

9.11 Photosynthesis

Microorganisms cannot only derive energy from the oxidation of inorganic and organic compounds, but many can capture light energy and use it to synthesize ATP and NADH or NADPH (figure 9.1; see also figure 8.8). This process in which light energy is trapped and converted to chemical energy is called **photosynthesis**. Usually a photosynthetic organism reduces and incorporates CO_2 , reactions that are also considered part of this process. Photosynthesis is one of the most significant metabolic processes on earth because almost all our energy is ultimately derived from solar energy. It provides photosynthetic organisms with the ATP and NADPH necessary to synthesize the organic material required for growth. In turn these organisms serve as the base of most food chains in the biosphere. Photosynthesis is also responsible for replenishing our supply of O_2 , a remarkable process carried out by a variety of organisms, both eucaryotic and procaryotic (table 9.6). Although most people associate photosynthesis with the more obvious higher plants, over half the photosynthesis on earth is carried out by microorganisms.

Photosynthesis as a whole is divided into two parts. In the **light reactions** light energy is trapped and converted to chemical energy. This energy is then used to reduce or fix CO_2 and synthesize cell constituents in the **dark reactions**. In this section the nature of the light reactions is discussed; the dark reactions are reviewed in the next chapter. [The photosynthetic dark reactions \(pp. 207–8\)](#)

Table 9.6 Diversity of Photosynthetic Organisms

Eucaryotic Organisms	Prokaryotic Organisms
Higher plants	Cyanobacteria (blue-green algae)
Multicellular green, brown, and red algae	Green sulfur bacteria
Unicellular algae (e.g., euglenoids, dinoflagellates, diatoms)	Green nonsulfur bacteria
	Purple sulfur bacteria
	Purple nonsulfur bacteria
	<i>Prochloron</i>

The Light Reaction in Eucaryotes and Cyanobacteria

All photosynthetic organisms have pigments for the absorption of light. The most important of these pigments are the **chlorophylls**. Chlorophylls are large planar rings composed of four substituted pyrrole rings with a magnesium atom coordinated to the four central nitrogen atoms (**figure 9.26**). Several chlorophylls are found in eucaryotes, the two most important of which are chlorophyll *a* and chlorophyll *b* (**figure 9.26**). These two molecules differ slightly in their structure and spectral properties. When dissolved in acetone, chlorophyll *a* has a light absorption peak at 665 nm; the corresponding peak for chlorophyll *b* is at 645 nm. In addition to absorbing red light, chlorophylls also absorb blue light strongly (the second absorption peak for chlorophyll *a* is at 430 nm). Because chlorophylls absorb primarily in the red and blue ranges, green light is transmitted. Consequently many photosynthetic organisms are green in color. The long hydrophobic tail attached to the chlorophyll ring aids in its attachment to membranes, the site of the light reactions.

Other photosynthetic pigments also trap light energy. The most widespread of these are the **carotenoids**, long molecules, usually yellowish in color, that possess an extensive conjugated double bond system (**figure 9.27**). β -Carotene is present in *Prochloron* and most divisions of algae; fucoxanthin is found in diatoms, dinoflagellates, and brown algae (*Phaeophyta*). Red algae and cyanobacteria have photosynthetic pigments called **phycobiliproteins**, consisting of a protein with a tetrapyrrole attached (**figure 9.27**). **Phycocerythrin** is a red pigment with a maximum absorption around 550 nm, and **phycocyanin** is blue (maximum absorption at 620 to 640 nm).

Carotenoids and phycobiliproteins are often called **accessory pigments** because of their role in photosynthesis. Although chlorophylls cannot absorb light energy effectively in the blue-green through yellow range (about 470 to 630 nm), accessory pigments do absorb light in this region and transfer the trapped energy to chlorophyll. In this way they make photosynthesis more efficient over a broader range of wavelengths. Accessory pigments also protect microorganisms from intense sunlight, which could oxidize and damage the photosynthetic apparatus in their absence.

Chlorophylls and accessory pigments are assembled in highly organized arrays called **antennas**, whose purpose is to create a large surface area to trap as many photons as possible. An antenna has about 300 chlorophyll molecules. Light energy is captured in an antenna and transferred from chlorophyll to

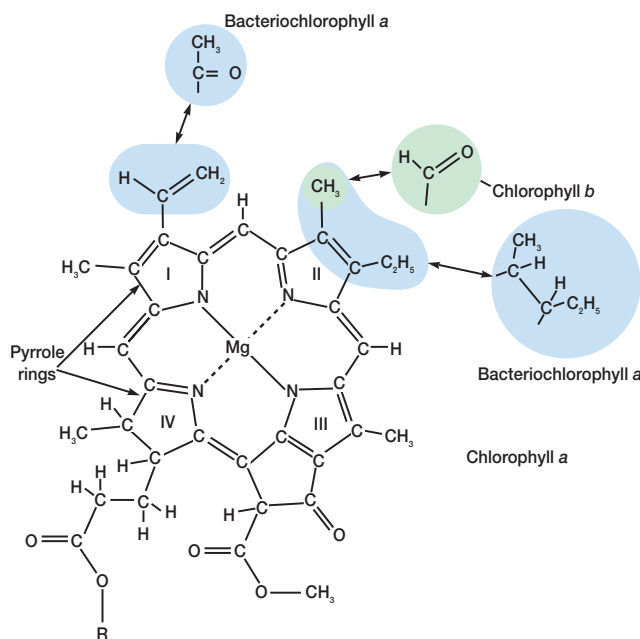


Figure 9.26 Chlorophyll Structure. The structures of chlorophyll *a*, chlorophyll *b*, and bacteriochlorophyll *a*. The complete structure of chlorophyll *a* is given. Only one group is altered to produce chlorophyll *b*, and two modifications in the ring system are required to change chlorophyll *a* to bacteriochlorophyll *a*. The side chain (R) of bacteriochlorophyll *a* may be either phytyl (a 20-carbon chain also found in chlorophylls *a* and *b*) or geranylgeranyl (a 20-carbon side chain similar to phytyl, but with three more double bonds).

chlorophyll until it reaches a special **reaction-center chlorophyll** directly involved in photosynthetic electron transport (**figure 9.28**). In eucaryotic cells and cyanobacteria, there are two kinds of antennas associated with two different photosystems. **Photosystem I** absorbs longer wavelength light (≥ 680 nm) and funnels the energy to a special chlorophyll *a* molecule called P700. The term P700 signifies that this molecule most effectively absorbs light at a wavelength of 700 nm. **Photosystem II** traps light at shorter wavelengths (≤ 680 nm) and transfers its energy to the special chlorophyll P680.

When the photosystem I antenna transfers light energy to the reaction-center P700 chlorophyll, P700 absorbs the energy and is excited; its reduction potential becomes very negative. It then donates its excited or high-energy electron to a specific acceptor, probably a special chlorophyll *a* molecule (A) or an iron-sulfur protein (**figure 9.29**). The electron is eventually transferred to ferredoxin and can then travel in either of two directions. In the cyclic pathway (the dashed lines in **figure 9.29**), the electron moves in a cyclic route through a series of electron carriers and back to the oxidized P700. The pathway is termed cyclic because the electron from P700 returns to P700 after traveling through the photosynthetic electron transport chain. PMF (p. 187) is formed during cyclic electron transport in the region of cytochrome *b₆*

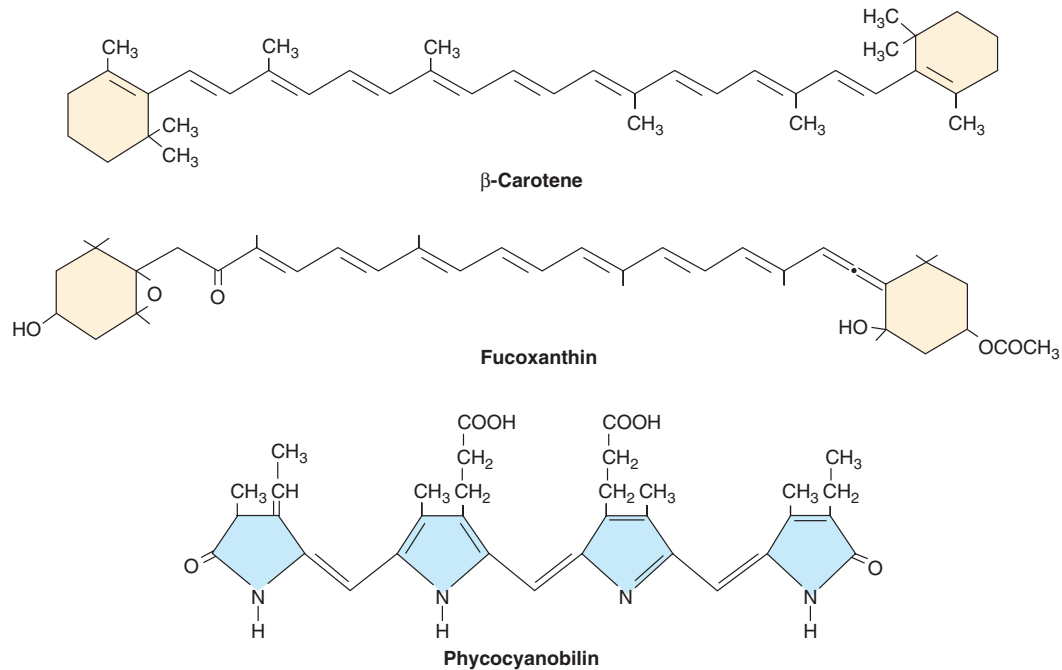


Figure 9.27 Representative Accessory Pigments. Beta-carotene is a carotenoid found in algae and higher plants. Note that it has a long chain of alternating double and single bonds called conjugated double bonds. Fucoxanthin is a carotenoid accessory pigment in several divisions of algae (the dot in the structure represents a carbon atom). Phycocyanobilin is an example of a linear tetrapyrrole that is attached to a protein to form a phycobiliprotein.

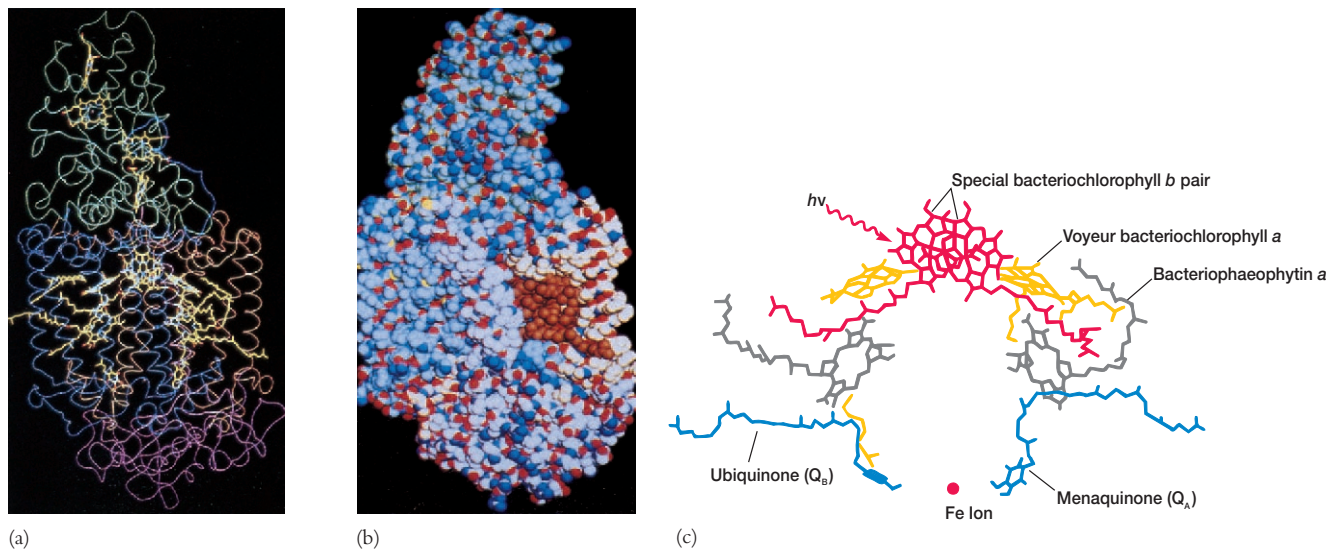


Figure 9.28 A Photosynthetic Reaction Chain. The reaction center of the purple nonsulfur bacterium, *Rhodospseudomonas viridis*. **(a)** The structure of the C_α backbone of the center's polypeptide chains with the bacteriochlorophylls and other prosthetic groups in yellow. **(b)** A space-filling model of the reaction center. The nitrogen atoms are blue; oxygen atoms, red; and sulfur atoms, yellow. The exposed prosthetic group atoms are reddish-brown. **(c)** A close-up view of the reaction center prosthetic groups. A photon is first absorbed by the "special pair" of bacteriochlorophyll *a* molecules, thus exciting them. An excited electron then moves to the bacteriopheophytin molecule in the right arm of the system.

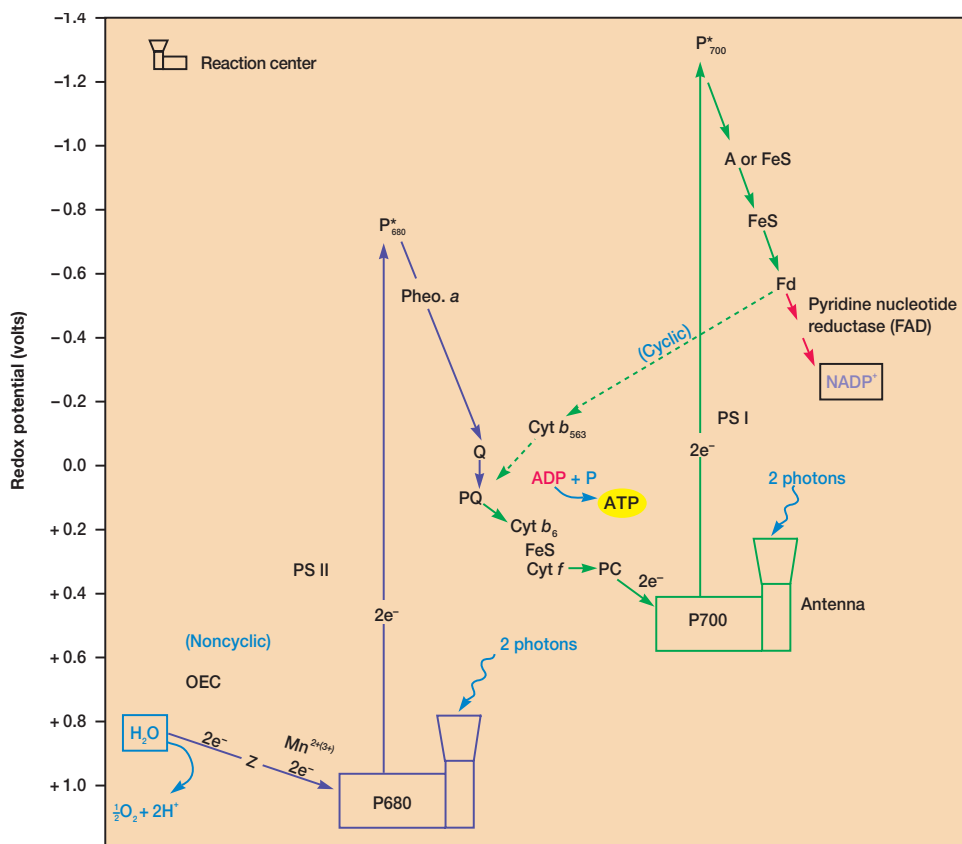


Figure 9.29 Green Plant Photosynthesis. Electron flow during photosynthesis in higher plants. Cyanobacteria and eucaryotic algae are similar in having two photosystems, although they may differ in some details. The carriers involved in electron transport are ferredoxin (Fd) and other FeS proteins; cytochromes b_6 , b_{563} , and f ; plastoquinone (PQ); copper containing plastocyanin (PC); pheophytin a (Pheo. a); possibly chlorophyll a (A); and the unknown quinone Q, which is probably a plastoquinone. Both photosystem I (PS I) and photosystem II (PS II) are involved in noncyclic photophosphorylation; only PS I participates in cyclic photophosphorylation. The oxygen evolving complex (OEC) that extracts electrons from water contains manganese ions and the substance Z, which transfers electrons to the PS II reaction center. See the text for further details.

and used to synthesize ATP. This process is called **cyclic photophosphorylation** because electrons travel in a cyclic pathway and ATP is formed. Only photosystem I participates.

Electrons also can travel in a noncyclic pathway involving both photosystems. P700 is excited and donates electrons to ferredoxin as before. In the noncyclic route, however, reduced ferredoxin reduces NADP^+ to NADPH (figure 9.29). Because the electrons contributed to NADP^+ cannot be used to reduce oxidized P700, photosystem II participation is required. It donates electrons to oxidized P700 and generates ATP in the process. The photosystem II antenna absorbs light energy and excites P680, which then reduces pheophytin a . Pheophytin a is chlorophyll a in which two hydrogen atoms have replaced the central magnesium. Electrons subsequently travel to Q (probably a plastoquinone) and down the electron transport chain to P700. Oxidized P680 then obtains an electron from the oxidation of water to O_2 .

Thus electrons flow from water all the way to NADP^+ with the aid of energy from two photosystems, and ATP is synthesized by **noncyclic photophosphorylation**. It appears that one ATP and one NADPH are formed when two electrons travel through the noncyclic pathway.

Just as is true of mitochondrial electron transport, photosynthetic electron transport takes place within a membrane. Chloroplast granal membranes contain both photosystems and their antennas. **Figure 9.30** shows a thylakoid membrane carrying out noncyclic photophosphorylation by the chemiosmotic mechanism. Protons move to the thylakoid interior during photosynthetic electron transport and return to the stroma when ATP is formed. It is believed that stromal lamellae possess only photosystem I and are involved in cyclic photophosphorylation alone. In cyanobacteria, photosynthetic light reactions are also located in membranes.

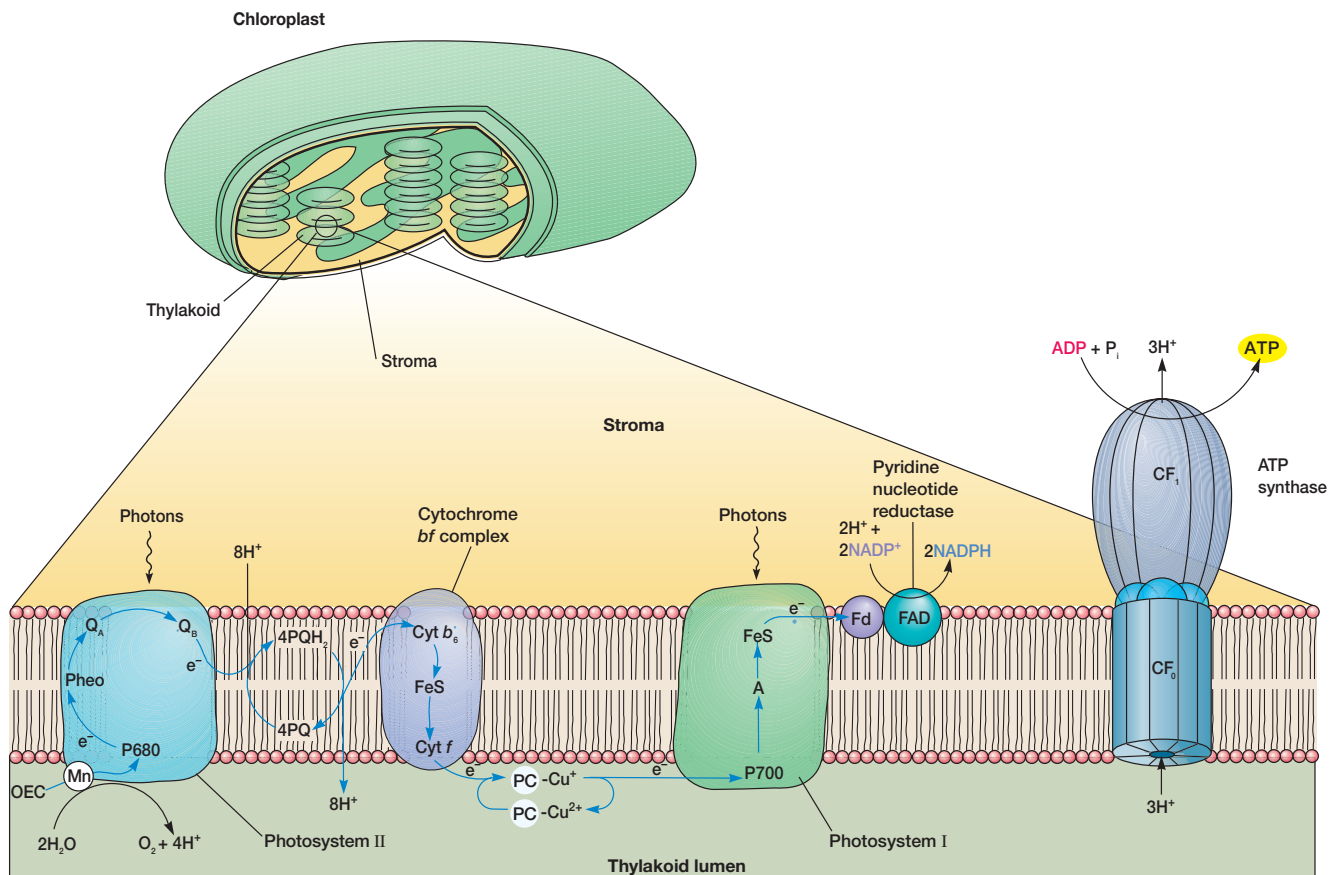
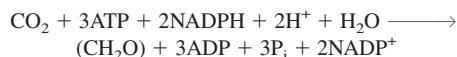


Figure 9.30 The Mechanism of Photosynthesis. An illustration of the chloroplast thylakoid membrane showing photosynthetic electron transport chain function and noncyclic photophosphorylation. The chain is composed of three complexes: PS I, the cytochrome *bf* complex, and PS II. Two diffusible electron carriers connect the three complexes. Plastoquinone (PQ) connects PS I with the cytochrome *bf* complex, and plastocyanin (PC) connects the cytochrome *bf* complex with PS II. The light-driven electron flow pumps protons across the thylakoid membrane and generates an electrochemical gradient, which can then be used to make ATP. Water is the source of electrons and the oxygen-evolving complex (OEC) produces oxygen.

The dark reactions require three ATPs and two NADPHs to reduce one CO₂ and use it to synthesize carbohydrate (CH₂O).



The noncyclic system generates one NADPH and one ATP per pair of electrons; therefore four electrons passing through the system will produce two NADPHs and two ATPs. A total of 8 quanta of light energy (4 quanta for each photosystem) is needed to propel the four electrons from water to NADP⁺. Because the ratio of ATP to NADPH required for CO₂ fixation is 3:2, at least one more ATP must be supplied. Cyclic photophosphorylation probably operates independently to generate the extra ATP. This requires absorption of another 2 to 4 quanta. It follows that around 10 to 12 quanta of light energy are needed to reduce and incorporate one molecule of CO₂ during photosynthesis.

The Light Reaction in Green and Purple Bacteria

Green and purple photosynthetic bacteria differ from cyanobacteria and eucaryotic photosynthesizers in several fundamental ways (table 9.7). In particular, green and purple bacteria do not use water as an electron source or produce O₂ photosynthetically—that is, they are **anoxygenic**. In contrast, cyanobacteria and eucaryotic photosynthesizers are almost always **oxygenic**. NADPH is not directly produced in the photosynthetic light reaction of purple bacteria. Green bacteria can reduce NAD⁺ directly during the light reaction. To synthesize NADH and NADPH, green and purple bacteria must use electron donors like hydrogen, hydrogen sulfide, elemental sulfur, and organic compounds that have more negative reduction potentials than water and are therefore easier to oxidize (better electron donors). Finally, green and purple bacteria possess slightly different photosynthetic pigments, **bacteriochlorophylls**.

Table 9.7 Properties of Microbial Photosynthetic Systems

Property	Eucaryotes	Cyanobacteria	Green and Purple Bacteria
Photosynthetic pigment	Chlorophyll <i>a</i>	Chlorophyll <i>a</i>	Bacteriochlorophyll
Photosystem II	Present	Present	Absent
Photosynthetic electron donors	H ₂ O	H ₂ O	H ₂ , H ₂ S, S, organic matter
O ₂ production pattern	Oxygenic	Oxygenic ^a	Anoxygenic
Primary products of energy conversion	ATP + NADPH	ATP + NADPH	ATP
Carbon source	CO ₂	CO ₂	Organic and/or CO ₂

^aSome cyanobacteria can function anoxygenically under certain conditions. For example, *Oscillatoria* can use H₂S as an electron donor instead of H₂O.

(figure 9.26), many with absorption maxima at longer wavelengths. Bacteriochlorophylls *a* and *b* have maxima in ether at 775 and 790 nm, respectively. In vivo maxima are about 830 to 890 nm (bacteriochlorophyll *a*) and 1,020 to 1,040 nm (Bchl *b*). This shift of absorption maxima into the infrared region better adapts these bacteria to their ecological niches (see figure 21.3). [The biology of photosynthetic bacteria](#) (pp. 468–76, 487–88, 498–501)

There are four groups of green and purple photosynthetic bacteria, each containing several genera: green sulfur bacteria (*Chlorobium*), green nonsulfur bacteria (*Chloroflexus*), purple sulfur bacteria (*Chromatium*), and purple nonsulfur bacteria (*Rhodospirillum*, *Rhodopseudomonas*). The biology of these groups will be discussed later.

Many differences found in green and purple bacteria are due to their lack of photosystem II; they cannot use water as an electron donor in noncyclic electron transport. Without photosystem II they cannot produce O₂ from H₂O photosynthetically and are restricted to cyclic photophosphorylation. Indeed, almost all purple and green sulfur bacteria are strict anaerobes. A tentative scheme for the photosynthetic electron transport chain of a purple nonsulfur bacterium is given in **figure 9.31**. When the special reaction-center chlorophyll P870 is excited, it donates an electron to bacteriopheophytin. Electrons then flow to quinones and through an electron transport chain back to P870 while driving ATP synthesis. Note that although both green and purple bacteria lack two photosystems, the purple bacteria have a photosynthetic apparatus similar to photosystem II, whereas the green sulfur bacteria have a system similar to photosystem I.

Green and purple bacteria face a further problem because they also require NADH or NADPH for CO₂ incorporation. They may synthesize NADH in at least three ways. If they are growing in the presence of hydrogen gas, which has a reduction potential more negative than that of NAD⁺, the hydrogen can be used di-

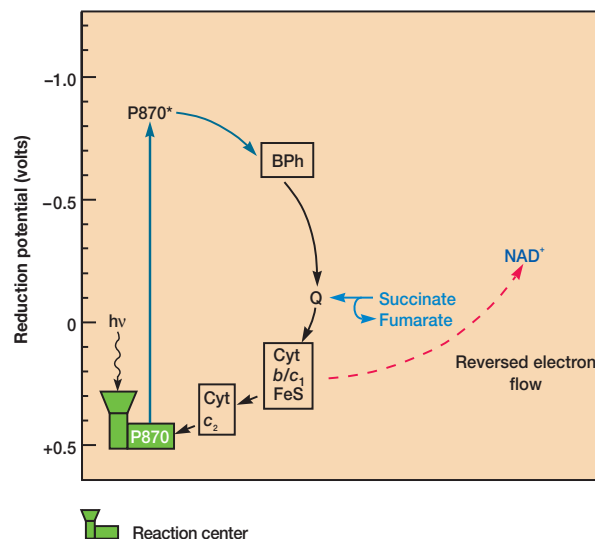


Figure 9.31 Purple Nonsulfur Bacterial Photosynthesis. The photosynthetic electron transport system in the purple nonsulfur bacterium, *Rhodobacter sphaeroides*. This scheme is incomplete and tentative. Ubiquinone (Q) is very similar to coenzyme Q. BPh stands for bacteriopheophytin. NAD⁺ and the electron source succinate are in color.

rectly to produce NADH. Like chemolithotrophs, many photosynthetic purple bacteria use proton motive force to reverse the flow of electrons in an electron transport chain and move them from inorganic or organic donors to NAD⁺ (**figures 9.31** and **9.32**). Green sulfur bacteria such as *Chlorobium* appear to carry out a simple form of noncyclic photosynthetic electron flow to reduce NAD⁺ (**figure 9.33**).

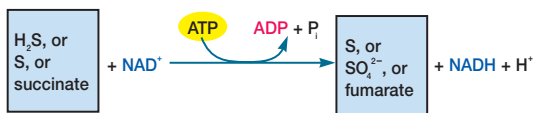


Figure 9.32 NAD Reduction in Green and Purple Bacteria. The use of reversed electron flow to reduce NAD^+ . The arrow in this diagram represents an electron transport chain that is being driven in reverse by proton motive force or ATP. That is, electrons are moving from donors with more positive reduction potentials to an acceptor (NAD^+) with a more negative potential.

1. Describe photosynthesis as carried out by eucaryotes and cyanobacteria. How does photosynthesis in green and purple bacteria differ?
2. Define the following terms: light reaction, chlorophyll, carotenoid, phycobiliprotein, accessory pigment, antenna, photosystems I and II, cyclic photophosphorylation, noncyclic photophosphorylation, anoxygenic and oxygenic photosynthesis, and bacteriochlorophyll.

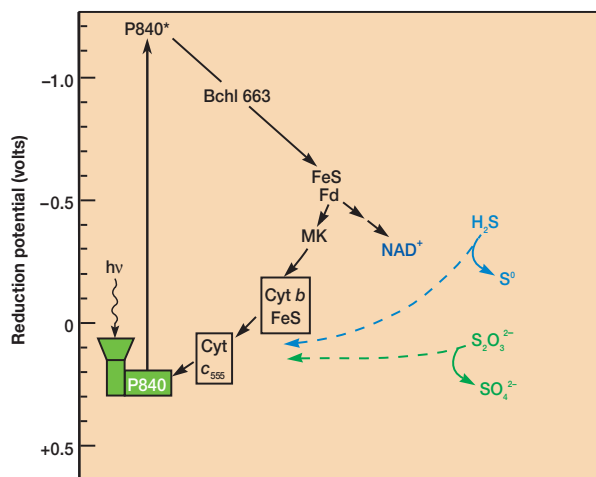


Figure 9.33 Green Sulfur Bacterial Photosynthesis. The photosynthetic electron transport system in the green sulfur bacterium, *Chlorobium limicola*. Light energy is used to make ATP by cyclic photophosphorylation and move electrons from sulfur donors (green and blue) to NAD^+ (red). The electron transport chain has a quinone called menaquinone (MK).

Summary

1. Metabolism is the total of all chemical reactions in a cell and may be divided into catabolism and anabolism.
2. Chemotrophic microorganisms can use three kinds of electron acceptors during energy metabolism (figure 9.2). The nutrient may be oxidized with an endogenous electron acceptor (fermentation), with oxygen as an exogenous electron acceptor (aerobic respiration), or with another external electron acceptor (anaerobic respiration).
3. Amphibolic pathways like glycolysis and the tricarboxylic acid cycle have both catabolic and anabolic functions.
4. The glycolytic or Embden-Meyerhof pathway (figure 9.5) occurs by way of fructose 1,6-bisphosphate with the net production of two NADHs and two ATPs, the latter being produced by substrate-level phosphorylation.
5. In the pentose phosphate pathway, glucose 6-phosphate is oxidized twice and converted to pentoses and other sugars. It is a source of NADPH, ATP, pentoses, and tetroses.
6. In the Entner-Doudoroff pathway, glucose is oxidized to 6-phosphogluconate, which is then dehydrated and cleaved to pyruvate and glyceraldehyde 3-phosphate. The latter product can be oxidized by glycolytic enzymes to provide ATP and NADH.
7. In the absence of O_2 a microorganism often uses an oxidized endogenous organic molecule as an electron acceptor to reoxidize the NADH formed during glycolysis (fermentation).
8. The tricarboxylic acid cycle is the final stage of catabolism in most aerobic cells (figure 9.12). It oxidizes acetyl-CoA to CO_2 and forms one GTP, three NADHs, and one FADH_2 per acetyl-CoA.
9. The NADH and FADH_2 produced from the oxidation of carbohydrates, fatty acids, and other nutrients can be oxidized in the electron transport chain. Electrons flow from carriers with more negative reduction potentials to those with more positive potentials (figure 9.13; see also figure 8.7), and free energy is released for ATP synthesis by oxidative phosphorylation.
10. Bacterial electron transport chains are often different from eucaryotic chains with respect to such aspects as carriers and branching. In eucaryotes the P/O ratio for NADH is about 3 and that for FADH_2 is around 2; P/O ratios are usually much lower in bacteria.
11. Eucaryotic ATP synthesis takes place on small protein spheres, the F_1F_0 ATPases or ATP synthases, located on the inner surface of the inner mitochondrial membrane. Bacterial ATP synthase is on the inner surface of the plasma membrane.
12. The most widely accepted mechanism of oxidative phosphorylation is the chemiosmotic hypothesis in which proton motive force drives ATP synthesis (figure 9.17).
13. When the glycolytic pathway operates anaerobically, only 2 ATPs per glucose are formed, whereas aerobic respiration in eucaryotes can yield a maximum of 38 ATPs.
14. Anaerobic respiration is the process of ATP production by electron transport in which the terminal electron acceptor is an exogenous, oxidized inorganic molecule other than O_2 . The most common acceptors are nitrate, sulfate, and CO_2 .
15. Microorganisms catabolize many extracellular carbohydrates. Monosaccharides are taken in and phosphorylated; disaccharides may be cleaved to monosaccharides by either hydrolysis or phosphorolysis.

202 Chapter 9 Metabolism: Energy Release and Conservation

16. External polysaccharides are degraded by hydrolysis and the products are absorbed. Intracellular glycogen and starch are converted to glucose 1-phosphate by phosphorylation.
17. Fatty acids from lipid catabolism are usually oxidized to acetyl-CoA in the β -oxidation pathway.
18. Proteins are hydrolyzed to amino acids that are then deaminated; their carbon skeletons feed into the TCA cycle.
19. Chemolithotrophs or chemoautotrophs synthesize ATP by oxidizing reduced inorganic compounds—usually hydrogen, reduced nitrogen and sulfur compounds, or ferrous iron—with an electron transport chain and O_2 as the electron acceptor.
20. Chemolithotrophs usually incorporate CO_2 through the Calvin cycle and produce NADH by reversing electron transport.
21. In photosynthesis, eucaryotes and cyanobacteria trap light energy with chlorophyll and accessory pigments and move electrons through photosystems I and II to make ATP and NADPH (the light reactions).
22. Cyclic photophosphorylation involves the activity of photosystem I alone. In noncyclic photophosphorylation photosystems I and II operate together to move electrons from water to $NADP^+$ producing ATP, NADPH, and O_2 (figure 9.29).
23. Photosynthetic green and purple bacteria differ from eucaryotes and cyanobacteria in possessing bacteriochlorophyll and lacking photosystem II (figures 9.31 and 9.33). Thus they cannot use the noncyclic pathway to form NADPH and O_2 ; they are anoxygenic.

Key Terms

- | | | |
|---------------------------------------|-------------------------------------|-------------------------------------|
| accessory pigments 196 | dark reactions 195 | oxidative phosphorylation 184 |
| acetyl coenzyme A (acetyl-CoA) 183 | deamination 192 | oxygenic photosynthesis 199 |
| adenosine 5'-phosphosulfate (APS) 194 | denitrification 190 | Pasteur effect 189 |
| aerobic respiration 173 | dissimilatory nitrate reduction 190 | pentose phosphate pathway 177 |
| alcoholic fermentation 179 | electron transport chain 184 | photosynthesis 195 |
| amphibolic pathways 176 | Embden-Meyerhof pathway 176 | photosystem I 196 |
| anabolism 173 | Entner-Doudoroff pathway 179 | photosystem II 196 |
| anaerobic respiration 173 | fermentation 173 | phycobiliproteins 196 |
| anoxygenic photosynthesis 199 | glycolysis 176 | phycocyanin 196 |
| antenna 196 | glycolytic pathway 176 | phycoerythrin 196 |
| ATP synthase 187 | heterolactic fermenters 181 | protease 192 |
| bacteriochlorophyll 199 | hexose monophosphate pathway 177 | proton motive force (PMF) 187 |
| β -oxidation pathway 192 | homolactic fermenters 181 | reaction-center chlorophyll 196 |
| butanediol fermentation 181 | Krebs cycle 183 | respiration 173 |
| carotenoids 196 | lactic acid fermentation 179 | Stickland reaction 181 |
| catabolism 173 | light reactions 195 | substrate-level phosphorylation 177 |
| chemiosmotic hypothesis 187 | metabolism 173 | transamination 192 |
| chemolithotroph 193 | mixed acid fermentation 181 | tricarboxylic acid (TCA) cycle 183 |
| chlorophylls 196 | nitrification 193 | uncouplers 189 |
| citric acid cycle 183 | nitrifying bacteria 193 | |
| cyclic photophosphorylation 198 | noncyclic photophosphorylation 198 | |

Questions for Thought and Review

1. Why might it be desirable for a microorganism with the Embden-Meyerhof pathway and the TCA cycle also to have the pentose phosphate pathway?
2. Of what advantage would it be to a microorganism to possess an electron transport chain and oxidative phosphorylation?
3. Describe two ways in which a microorganism can continue to produce energy when O_2 is absent.
4. Why would it be wasteful for anaerobic microorganisms to operate the complete TCA cycle?
5. How do substrate-level phosphorylation and oxidative phosphorylation differ from one another?
6. Can fermentation products be used in identifying bacteria? Give some examples if the answer is yes.
7. Describe what would happen to microbial metabolism if the enzyme lactate dehydrogenase was inhibited in a homolactic fermenter growing anaerobically on a medium containing glucose as the carbon source. What effects would an inhibitor of the ATP synthase have on an aerobically respiring microorganism? An uncoupler?
8. How would you isolate a thermophilic chemolithotroph that uses sulfur compounds as a source of electrons? What changes in the incubation system would be needed to isolate bacteria using sulfur compounds in anaerobic respiration? How can one tell which process is taking place through an analysis of the sulfur molecules present in the medium?
9. Suppose that you isolated a bacterial strain that carried out oxygenic photosynthesis. What photosystems would it possess and what group of bacteria would it most likely belong to?

Critical Thinking Questions

1. Without looking in chapter 21, predict some conditions that would describe niches occupied by green and purple photosynthetic bacteria.
2. From an evolutionary perspective, discuss why all microorganisms (with only rare exceptions) use aerobic respiration to generate ATP.

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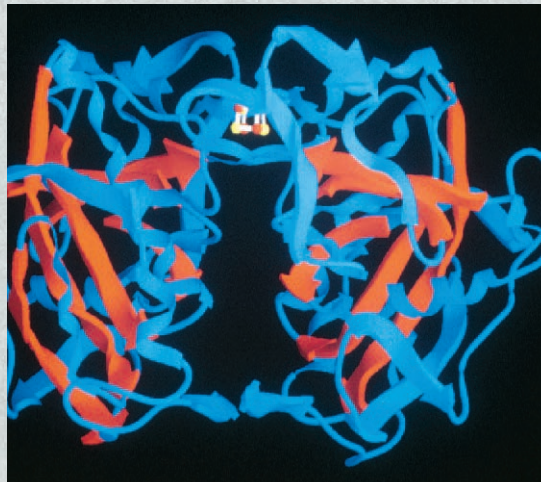
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CHAPTER 10

Metabolism: The Use of Energy in Biosynthesis



The nitrogenase Fe protein's subunits are arranged like a pair of butterfly wings. Nitrogenase consists of the Fe protein and the MoFe protein; it catalyzes the reduction of atmospheric nitrogen during nitrogen fixation.

Outline

10.1	Principles Governing Biosynthesis	205	10.5	The Synthesis of Amino Acids	214
10.2	The Photosynthetic Fixation of CO ₂	207	10.6	Anaplerotic Reactions	215
	The Carboxylation Phase	208	10.7	The Synthesis of Purines, Pyrimidines, and Nucleotides	216
	The Reduction Phase	208		Purine Biosynthesis	217
	The Regeneration Phase	208		Pyrimidine Biosynthesis	218
10.3	Synthesis of Sugars and Polysaccharides	209	10.8	Lipid Synthesis	218
10.4	The Assimilation of Inorganic Phosphorus, Sulfur, and Nitrogen	210	10.9	Peptidoglycan Synthesis	221
	Phosphorus Assimilation	210	10.10	Patterns of Cell Wall Formation	223
	Sulfur Assimilation	210			
	Nitrogen Assimilation	210			
	Nitrogen Fixation	212			

Concepts

1. In anabolism or biosynthesis, cells use free energy to construct more complex molecules and structures from smaller, simpler precursors.
2. Biosynthetic pathways are organized to optimize efficiency by conserving biosynthetic raw materials and energy.
3. Autotrophs use ATP and NADPH from photosynthesis or from oxidation of inorganic molecules to reduce CO_2 and incorporate it into organic material.
4. Catabolic and anabolic pathways may differ in enzymes, regulation, intracellular location, and use of cofactors and nucleoside diphosphate carriers. Although many enzymes of amphibolic pathways participate in both catabolism and anabolism, some pathway enzymes are involved only in one of the two processes.
5. Phosphorus, in the form of phosphate, can be directly assimilated, whereas inorganic sulfur and nitrogen compounds must often be reduced before incorporation into organic material.
6. The tricarboxylic acid (TCA) cycle acts as an amphibolic pathway and requires anaplerotic reactions to maintain adequate levels of cycle intermediates.
7. Most glycolytic enzymes participate in both the synthesis and catabolism of glucose. In contrast, fatty acids are synthesized from acetyl-CoA and malonyl-CoA by a pathway quite different from fatty acid β -oxidation.
8. Peptidoglycan synthesis is a complex, multistep process that is begun in the cytoplasm and completed at the cell wall after the peptidoglycan repeat unit has been transported across the plasma membrane.

Biological structures are almost always constructed in a hierarchical manner, with subassemblies acting as important intermediates en route from simple starting molecules to the end products of organelles, cells, and organisms.

—W. M. Becker and D. W. Deamer

As the last chapter makes clear, microorganisms can obtain energy in many ways. Much of this energy is used in biosynthesis or anabolism. During biosynthesis a microorganism begins with simple precursors, such as inorganic molecules and monomers, and constructs ever more complex molecules until new organelles and cells arise (**figure 10.1**). A microbial cell must manufacture many different kinds of molecules; however, it is possible to discuss the synthesis of only the most important types of cell constituents.

This chapter begins with a general introduction to anabolism, then focuses on the synthesis of carbohydrates, amino acids, purines and pyrimidines, and lipids. It also describes the assimilation of CO_2 , phosphorus, sulfur, and nitrogen. The chapter ends with a section on the synthesis of peptidoglycan and bacterial cell walls. Protein and nucleic acid synthesis is so significant and complex that it is described separately in chapters 11 and 12.

Because anabolism is the creation of order and a cell is highly ordered and immensely complex, much energy is required for biosynthesis. This is readily apparent from estimates of the biosynthetic capacity of rapidly growing *Escherichia coli* (**table 10.1**). Although most ATP dedicated to biosynthesis is employed in protein synthesis, ATP is also used to make other cell constituents.

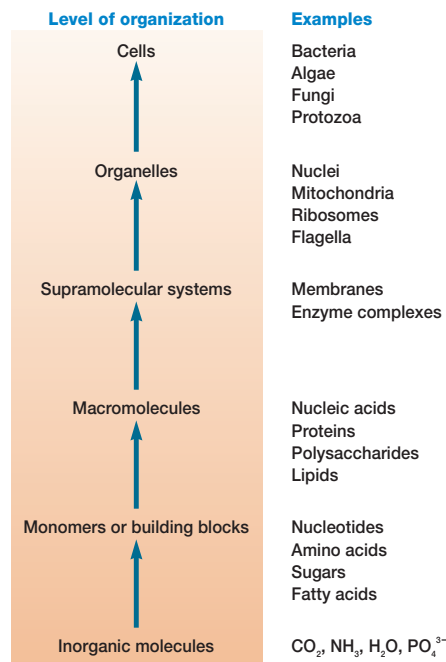


Figure 10.1 The Construction of Cells. The biosynthesis of procaryotic and eucaryotic cell constituents. Biosynthesis is organized in levels of ever greater complexity.

Free energy is required for biosynthesis in mature cells of constant size because cellular molecules are continuously being degraded and resynthesized, a process known as **turnover**. Cells are never the same from instant to instant. Despite the continuous turnover of cell constituents, metabolism is carefully regulated so that the rate of biosynthesis is approximately balanced by that of catabolism. In addition to the energy expended in the turnover of molecules, many nongrowing cells also use energy to synthesize enzymes and other substances for release into their surroundings.

[Regulation of metabolism \(chapters 8 and 12\)](#)

10.1 Principles Governing Biosynthesis

Biosynthetic metabolism seems to follow certain patterns or be shaped by a few general principles. Six of these are now briefly discussed.

1. A microbial cell contains large quantities of proteins, nucleic acids, and polysaccharides, all of which are **macromolecules** or very large molecules that are polymers of smaller units joined together. The construction of large, complex molecules from a few simple structural units or **monomers** saves much genetic storage capacity, biosynthetic raw material, and energy. A consideration of protein synthesis clarifies this. Proteins—whatever size, shape, or function—are made of only 20 common amino acids joined by peptide bonds (*see appendix I*). Different

Table 10.1 Biosynthesis in *Escherichia coli*

Cell Constituent	Number of Molecules per Cell ^a	Molecules Synthesized per Second	Molecules of ATP Required per Second for Synthesis
DNA	1 ^b	0.00083	60,000
RNA	15,000	12.5	75,000
Polysaccharides	39,000	32.5	65,000
Lipids	15,000,000	12,500.0	87,000
Proteins	1,700,000	1,400.0	2,120,000

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^aEstimates for a cell with a volume of $2.25 \mu\text{m}^3$, a total weight of $1 \times 10^{-12}\text{g}$, a dry weight of $2.5 \times 10^{-13}\text{g}$, and a 20 minute cell division cycle.

^bIt should be noted that bacteria can contain multiple copies of their genomic DNA.

proteins simply have different amino acid sequences but not new and dissimilar amino acids. Suppose that proteins were composed of 40 different amino acids instead of 20. The cell would then need the enzymes to manufacture twice as many amino acids (or would have to obtain the extra amino acids in its diet). Genes would be required for the extra enzymes, and the cell would have to invest raw materials and energy in the synthesis of these additional genes, enzymes, and amino acids. Clearly the use of a few monomers linked together by a single type of covalent bond makes the synthesis of macromolecules a highly efficient process. Almost all cell structures are built mainly of about 30 small precursors.

- The cell often saves additional materials and energy by using many of the same enzymes for both catabolism and anabolism. For example, most glycolytic enzymes are involved in the synthesis and the degradation of glucose.
- Although many enzymes in amphibolic pathways (*see section 9.1*) participate in both catabolic and anabolic activities, some steps are catalyzed by two different enzymes. One enzyme catalyzes the reaction in the catabolic direction, the other reverses this conversion (**figure 10.2**). Thus catabolic and anabolic pathways are never identical although many enzymes are shared. Use of separate enzymes for the two directions of a single step permits independent regulation of catabolism and anabolism. Although this has been discussed in more detail in sections 8.7 through 8.9, note that the regulation of anabolism is somewhat different from that of catabolism. Both types of pathways can be regulated by their end products as well as by the concentrations of ATP, ADP, AMP, and NAD^+ . Nevertheless, end product regulation generally assumes more importance in anabolic pathways.
- To synthesize molecules efficiently, anabolic pathways must operate irreversibly in the direction of biosynthesis. Cells can achieve this by connecting some biosynthetic reactions to the breakdown of ATP and other nucleoside triphosphates. When these two processes are coupled, the free energy made available during nucleoside triphosphate breakdown drives the biosynthetic reaction to completion (*see sections 8.3 and 8.4*).

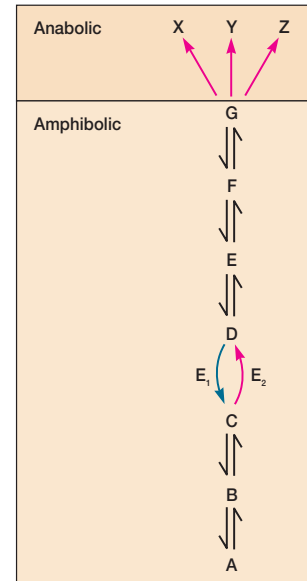


Figure 10.2 A Hypothetical Biosynthetic Pathway. The routes connecting G with X, Y, and Z are purely anabolic because they are used only for synthesis of the end products. The pathway from A to G is amphibolic—that is, it has both catabolic and anabolic functions. Most reactions are used in both roles; however, the interconversion of C and D is catalyzed by two separate enzymes, E_1 (catabolic) and E_2 (anabolic).

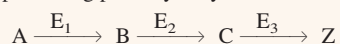
- In eucaryotic microorganisms biosynthetic pathways are frequently located in different cellular compartments from their corresponding catabolic pathways (**Box 10.1**). For example, fatty acid biosynthesis occurs in the cytoplasmic matrix, whereas fatty acid oxidation takes place within the mitochondrion. Compartmentation makes it easier for the pathways to operate simultaneously but independently.
- Finally, anabolic and catabolic pathways often use different cofactors. Usually catabolic oxidations produce NADH, a substrate for electron transport. In contrast, when a

Box 10.1

The Identification of Anabolic Pathways

There are three approaches to the study of pathway organization: (1) study of the pathway in vitro, (2) use of nutritional mutants, and (3) incubation of cells with precursors labeled radioisotopically. In vitro [Latin, in glass] studies employ cell-free extracts to search for enzymes and metabolic intermediates that might belong to a pathway. Although this direct approach was used to work out the organization of many catabolic pathways, progress in research on biosynthesis was slow until the other two techniques were developed in the early to middle 1940s.

Techniques using nutritional mutants were developed during Beadle and Tatum's work on the genetics of the red bread mold *Neurospora*. This approach is best illustrated with a hypothetical example. Suppose that a pathway for the synthesis of end product Z is organized with E₁, E₂, and so on, representing pathway enzymes.



The prototroph (see section 11.6), which will grow in medium lacking Z, can be treated with mutagenic agents such as ultraviolet light, X rays, or chemical mutagens. Some resulting mutants will be auxotrophs that require the presence of Z for growth because one of their biosynthetic enzymes is now inactive. When E₃ is inactive, the microorganism will grow only in the presence of Z, even though it can make C from the precursor A. When grown in the presence of a small amount of Z, intermediate C (the interme-

diate just before the blocked step) will accumulate in the medium. In this way a variety of mutants can be used to establish the identity of pathway intermediates. The order of intermediates can be determined by cross-feeding experiments. If E₂ has been inactivated by a mutation, the mutant will grow only when either C or Z is supplied. Because the medium in which the E₃ mutant has been cultured contains intermediate C, it will support growth of the E₂ mutant (other mutants would not produce enough C to support growth). If cross-feeding experiments are conducted with mutants of each step in the pathway, the steps can be placed in the correct order. Application of this technique quickly led to the elucidation of the pathways for the synthesis of tryptophan, folic acid, and other molecules.

Radioisotopes such as ¹⁴C are used in the third approach to studying pathway organization. Potential biosynthetic precursors are synthesized in the laboratory with specific atoms made radioactive. The microorganism then is incubated with culture medium containing the radioactive molecule, and the biosynthetic end product is isolated and analyzed. If the molecule truly is a precursor of the end product, the latter should be radioactive. The location of the radioactive atom will determine what part of the product is contributed by the radioactively labeled precursor. Precisely the same approach can be employed with nonradioactive atoms like ¹⁵N. This technique provided some of the first information about the nature of the purine biosynthetic pathway.

reductant is needed during biosynthesis, NADPH rather than NADH normally serves as the donor. Fatty acid metabolism provides a second example. Fatty acyl-CoA molecules are oxidized to generate energy, whereas fatty acid synthesis involves acyl carrier protein thioesters (p. 220).

After macromolecules have been constructed from simpler precursors, they are assembled into larger, more complex structures such as supramolecular systems and organelles (figure 10.1). Macromolecules normally contain the necessary information to form spontaneously in a process known as **self-assembly**. For example, ribosomes are large assemblages of many proteins and ribonucleic acid molecules, yet they arise by the self-assembly of their components without the involvement of extra factors.

1. Define biosynthesis or anabolism and turnover.
2. List six principles by which biosynthetic pathways are organized.

10.2 The Photosynthetic Fixation of CO₂

Although most microorganisms can incorporate or fix CO₂, at least in anaplerotic reactions (pp. 215–16), only autotrophs use CO₂ as their sole or principal carbon source. The reduction and incorporation of CO₂ requires much energy. Usually autotrophs obtain energy by trapping light during photosynthesis, but some

derive energy from the oxidation of reduced inorganic electron donors. Autotrophic CO₂ fixation is crucial to life on earth because it provides the organic matter on which heterotrophs depend. [Photosynthetic light reactions and chemolithotrophy \(pp. 193–201\)](#)

Microorganisms can fix CO₂ or convert this inorganic molecule to organic carbon and assimilate it in three major ways. Almost all microbial autotrophs incorporate CO₂ by a special metabolic pathway called by several names: the **Calvin cycle**, Calvin-Benson cycle, or reductive pentose phosphate cycle. Although the Calvin cycle is found in photosynthetic eucaryotes and most photosynthetic procaryotes, it is absent in the Archaea, some obligately anaerobic bacteria, and some microaerophilic bacteria. These microorganisms usually employ one of two other pathways. A reductive tricarboxylic acid pathway (see figure 20.6a) is used by some archaea (*Thermoproteus*, *Sulfolobus*) and by bacteria such as *Chlorobium* and *Desulfobacter*. The acetyl-CoA pathway (see figure 20.6b) is found in methanogens, sulfate reducers, and bacteria that can form acetate from CO₂ during fermentation (acetogens). Because of its importance, we will focus on the Calvin cycle here. [Alternate CO₂ fixation pathways \(pp. 454–55\)](#)

The Calvin cycle is found in the chloroplast stroma of eucaryotic microbial autotrophs. Cyanobacteria, some nitrifying bacteria, and thiobacilli possess **carboxysomes**. These are polyhedral inclusion bodies that contain the enzyme ribulose-1,5-bisphosphate carboxylase (see following section). They may be the site of CO₂ fixation or may store the carboxylase and other proteins. Understanding the cycle is easiest if it is divided into three phases: carboxylation,

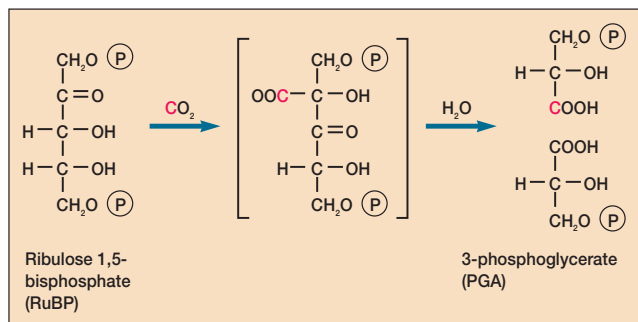


Figure 10.3 The Ribulose-1,5-Bisphosphate Carboxylase Reaction. This enzyme catalyzes the addition of carbon dioxide to ribulose 1,5-bisphosphate, forming an unstable intermediate, which then breaks down to two molecules of 3-phosphoglycerate.

reduction, and regeneration. An overview of the cycle is given in figure 10.4 and the details are presented in appendix II.

The Carboxylation Phase

Carbon dioxide fixation is accomplished by the enzyme **ribulose 1,5-bisphosphate carboxylase** or ribulosebisphosphate carboxylase/oxygenase (rubisco) (**figure 10.3**), which catalyzes the addition of CO_2 to ribulose 1,5-bisphosphate (RuBP), forming two molecules of 3-phosphoglycerate (PGA).

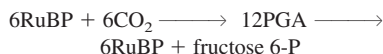
The Reduction Phase

After PGA is formed by carboxylation, it is reduced to glyceraldehyde 3-phosphate. The reduction, carried out by two enzymes, is essentially a reversal of a portion of the glycolytic pathway, although the glyceraldehyde 3-phosphate dehydrogenase differs from the glycolytic enzyme in using NADP^+ rather than NAD^+ (**figure 10.4**).

The Regeneration Phase

The third phase of the Calvin cycle regenerates RuBP and produces carbohydrates such as glyceraldehyde 3-phosphate, fructose, and glucose (**figure 10.4**). This portion of the cycle is similar to the pentose phosphate pathway and involves the transketolase and transaldolase reactions. The cycle is completed when phosphoribulokinase reforms RuBP.

To synthesize fructose 6-phosphate or glucose 6-phosphate from CO_2 , the cycle must operate six times to yield the desired hexose and reform the six RuBP molecules.



The incorporation of one CO_2 into organic material requires three ATPs and two NADPHs. The formation of glucose from CO_2 may be summarized by the following equation.

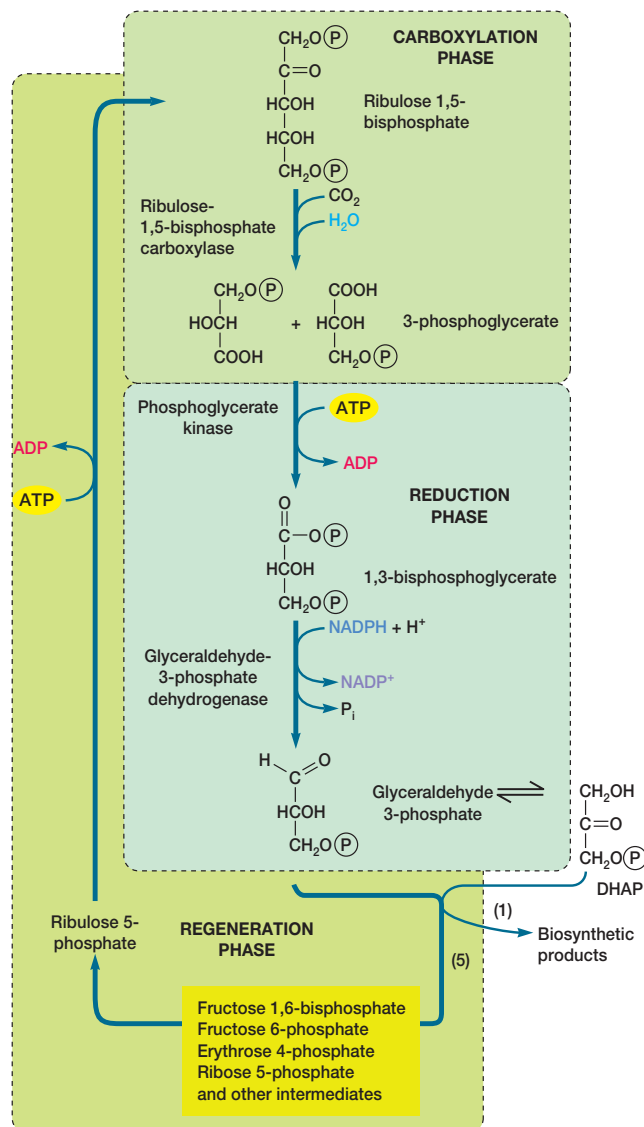
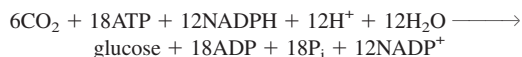


Figure 10.4 The Calvin Cycle. This is an overview of the cycle with only the carboxylation and reduction phases in detail. Three ribulose 1,5-bisphosphates are carboxylated to give six 3-phosphoglycerates in the carboxylation phase. These are converted to six glyceraldehyde 3-phosphates, which can be converted to dihydroxyacetone phosphate (DHAP). Five of the six trioses (glyceraldehyde phosphate and dihydroxyacetone phosphate) are used to reform three ribulose 1,5-bisphosphates in the regeneration phase. The remaining triose is used in biosynthesis.

ATP and NADPH are provided by photosynthetic light reactions or by oxidation of inorganic molecules in chemoautotrophs. Sugars formed in the Calvin cycle can then be used to synthesize other essential molecules.

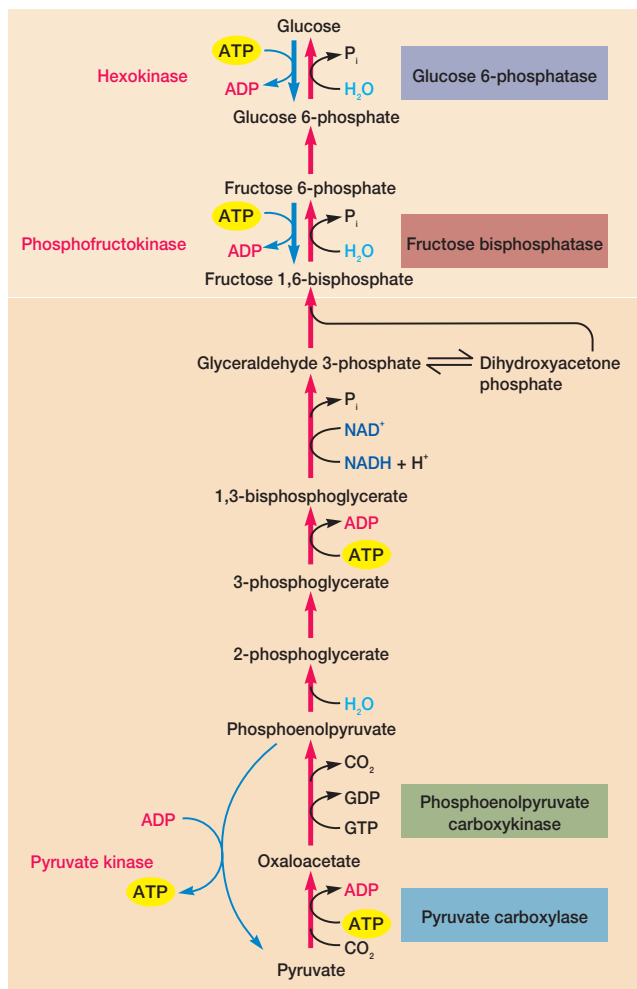


Figure 10.5 Gluconeogenesis. The gluconeogenic pathway used in many microorganisms. The names of the four enzymes catalyzing reactions different from those found in glycolysis are in shaded boxes. Glycolytic steps are shown in blue for comparison.

10.3 Synthesis of Sugars and Polysaccharides

Many microorganisms cannot carry out photosynthesis and are heterotrophs that must synthesize sugars from reduced organic molecules rather than from CO_2 . The synthesis of glucose from noncarbohydrate precursors is called **gluconeogenesis**. Although the gluconeogenic pathway is not identical with the glycolytic pathway, they do share seven enzymes (**figure 10.5**). Three glycolytic steps are irreversible in the cell: (1) the conversion of phosphoenolpyruvate to pyruvate, (2) the formation of fructose 1,6-bisphosphate from fructose 6-phosphate, and (3) the phosphorylation of glucose. These must be bypassed when the pathway is operating biosynthetically. For example, the formation of fructose 1,6-bisphosphate by phosphofructokinase is reversed by the enzyme, fructose biphosphatase, which hydrolytically re-

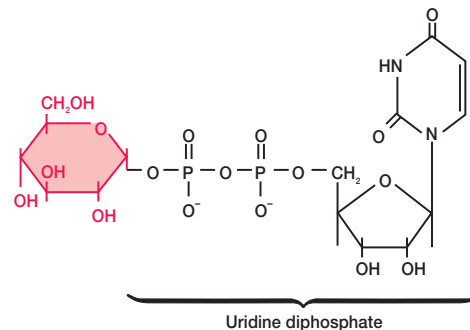


Figure 10.6 Uridine Diphosphate Glucose. Glucose is in color.

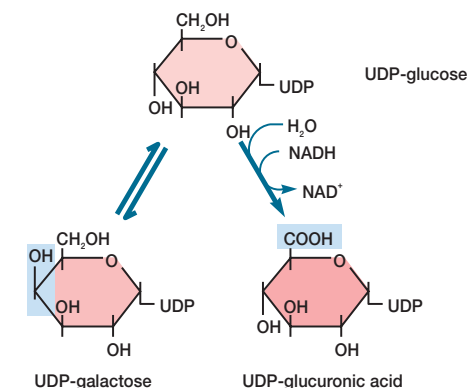
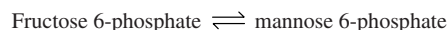


Figure 10.7 Uridine Diphosphate Galactose and Glucuronate Synthesis. The synthesis of UDP-galactose and UDP-glucuronic acid from UDP-glucose. Structural changes are indicated by colored boxes.

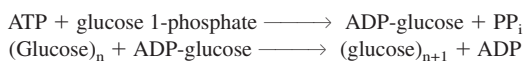
moves a phosphate from fructose bisphosphate. Usually at least two enzymes are involved in the conversion of pyruvate to phosphoenolpyruvate (the reversal of the pyruvate kinase step).

As can be seen in **figure 10.5**, the pathway synthesizes fructose as well as glucose. Once glucose and fructose have been formed, other common sugars can be manufactured. For example, mannose comes directly from fructose by a simple rearrangement.



Several sugars are synthesized while attached to a nucleoside diphosphate. The most important nucleoside diphosphate sugar is **uridine diphosphate glucose (UDPG)**. Glucose is activated by attachment to the pyrophosphate of uridine diphosphate through a reaction with uridine triphosphate (**figure 10.6**). The UDP portion of UDPG is recognized by enzymes and carries glucose around the cell for participation in enzyme reactions much like ADP bears phosphate in the form of ATP. UDP-galactose is synthesized from UDPG through a rearrangement of one hydroxyl group. A different enzyme catalyzes the synthesis of UDP-glucuronic acid through the oxidation of UDPG (**figure 10.7**).

Nucleoside diphosphate sugars also play a central role in the synthesis of polysaccharides such as starch and glycogen. Again, biosynthesis is not simply a direct reversal of catabolism. Glycogen and starch catabolism (*see section 9.7*) proceeds either by hydrolysis to form free sugars or by the addition of phosphate to these polymers with the production of glucose 1-phosphate. Nucleoside diphosphate sugars are not involved. In contrast, during the synthesis of glycogen and starch in bacteria and algae, adenosine diphosphate glucose is formed from glucose 1-phosphate and then donates glucose to the end of growing glycogen and starch chains.



Nucleoside diphosphate sugars also participate in the synthesis of complex molecules such as bacterial cell walls (pp. 221–23).

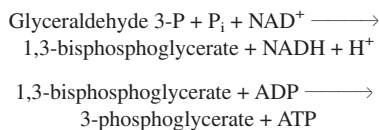
1. Briefly describe the three stages of the Calvin cycle.
2. What is gluconeogenesis and how does it usually occur? Describe the formation of mannose, galactose, starch, and glycogen. Why are nucleoside diphosphate sugars important?

10.4 The Assimilation of Inorganic Phosphorus, Sulfur, and Nitrogen

Besides carbon and oxygen, microorganisms also require large quantities of phosphorus, sulfur, and nitrogen for biosynthesis. Each of these is assimilated, or incorporated into organic molecules, by different routes. [Microbial nutrition \(chapter 5\)](#); [Microbial participation in biogeochemical cycles \(section 28.4\)](#)

Phosphorus Assimilation

Phosphorus is found in nucleic acids, proteins, phospholipids, ATP, and coenzymes like NADP. The most common phosphorus sources are inorganic phosphate and organic phosphate esters. Inorganic phosphate is incorporated through the formation of ATP in one of three ways: by (1) photophosphorylation (*see pp. 196–99*), (2) oxidative phosphorylation (*see pp. 187–89*), and (3) substrate-level phosphorylation. Glycolysis provides an example of the latter process. Phosphate is joined with glyceraldehyde 3-phosphate to give 1,3-bisphosphoglycerate, which is next used in ATP synthesis.



Microorganisms may obtain organic phosphates from their surroundings in dissolved or particulate form. **Phosphatases** very often hydrolyze organic phosphate esters to release inorganic phosphate. Gram-negative bacteria have phosphatases in the periplasmic space between their cell wall and the plasma membrane, which allows phosphate to be taken up immediately after release. On the other hand, protozoa can directly use organic

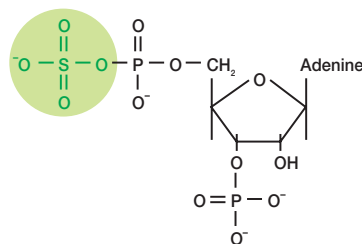


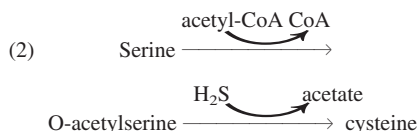
Figure 10.8 Phosphoadenosine 5'-phosphosulfate (PAPS). The sulfate group is in color.

phosphates after ingestion or hydrolyze them in lysosomes and incorporate the phosphate.

Sulfur Assimilation

Sulfur is needed for the synthesis of amino acids (cysteine and methionine) and several coenzymes (e.g., coenzyme A and biotin) and may be obtained from two sources. Many microorganisms use cysteine and methionine, obtained from either external sources or intracellular amino acid reserves. In addition, sulfate can provide sulfur for biosynthesis. The sulfur atom in sulfate is more oxidized than it is in cysteine and other organic molecules; thus sulfate must be reduced before it can be assimilated. This process is known as **assimilatory sulfate reduction** to distinguish it from the **dissimilatory sulfate reduction** that takes place when sulfate acts as an electron acceptor during anaerobic respiration (*see figure 28.21*). [Anaerobic respiration \(pp. 190–91\)](#)

Assimilatory sulfate reduction involves sulfate activation through the formation of **phosphoadenosine 5'-phosphosulfate** (**figure 10.8**), followed by reduction of the sulfate. The process is a complex one (**figure 10.9**) in which sulfate is first reduced to sulfite (SO_3^{2-}), then to hydrogen sulfide. Cysteine can be synthesized from hydrogen sulfide in two ways. Fungi appear to combine hydrogen sulfide with serine to form cysteine (process 1), whereas many bacteria join hydrogen sulfide with O-acetylserine instead (process 2).



Once formed, cysteine can be used in the synthesis of other sulfur-containing organic compounds.

Nitrogen Assimilation

Because nitrogen is a major component of proteins, nucleic acids, coenzymes, and many other cell constituents, the cell's ability to assimilate inorganic nitrogen is exceptionally important. Al-

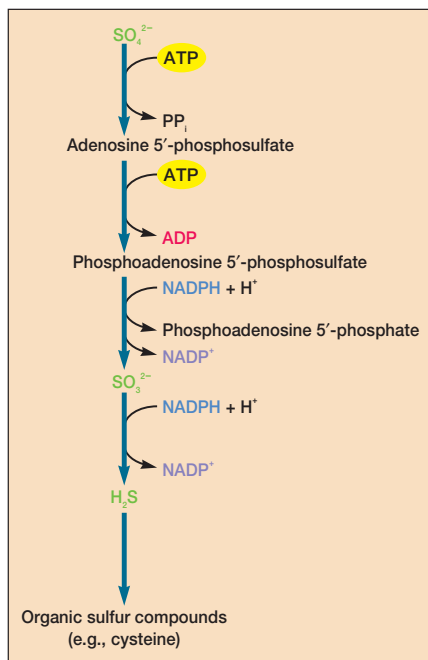


Figure 10.9 The Sulfate Reduction Pathway.

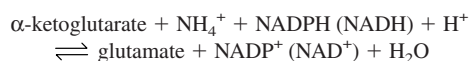
though nitrogen gas is abundant in the atmosphere, few microorganisms can reduce the gas and use it as a nitrogen source. Most must incorporate either ammonia or nitrate.

Ammonia Incorporation

Ammonia nitrogen can be incorporated into organic material relatively easily and directly because it is more reduced than other forms of inorganic nitrogen. Some microorganisms form the amino acid alanine in a reductive amination reaction catalyzed by alanine dehydrogenase.



The major route for ammonia incorporation often is the formation of glutamate from α -ketoglutarate (a TCA cycle intermediate). Many bacteria and fungi employ **glutamate dehydrogenase**, at least when the ammonia concentration is high.



Different species vary in their ability to use NADPH and NADH as the reducing agent in glutamate synthesis.

Once either alanine or glutamate has been synthesized, the newly formed α -amino group can be transferred to other carbon skeletons by transamination reactions (see section 9.9) to

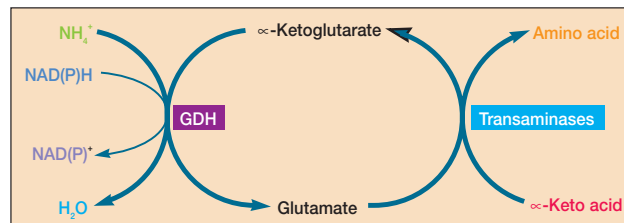


Figure 10.10 The Ammonia Assimilation Pathway. Ammonia assimilation by use of glutamate dehydrogenase (GDH) and transaminases. Either NADP- or NAD-dependent glutamate dehydrogenases may be involved. This route is most active at high ammonia concentrations.

form different amino acids. **Transaminases** possess the coenzyme pyridoxal phosphate, which is responsible for the amino group transfer. Microorganisms have a number of transaminases, each of which catalyzes the formation of several amino acids using the same amino acid as an amino group donor. When glutamate dehydrogenase works in cooperation with transaminases, ammonia can be incorporated into a variety of amino acids (figure 10.10).

A second route of ammonia incorporation involves two enzymes acting in sequence, **glutamine synthetase** and **glutamate synthase** (figure 10.11). Ammonia is used to synthesize glutamine from glutamate, then the amide nitrogen of glutamine is transferred to α -ketoglutarate to generate a new glutamate molecule. Because glutamate acts as an amino donor in transaminase reactions, ammonia may be used to synthesize all common amino acids when suitable transaminases are present (figure 10.12). Both ATP and a source of electrons, such as NADPH or reduced ferredoxin, are required. This route is present in *Escherichia coli*, *Bacillus megaterium*, and other bacteria. The two enzymes acting in sequence operate very effectively at low ammonia concentrations, unlike the glutamate dehydrogenase pathway. As we saw earlier, glutamine synthetase is tightly regulated by reversible covalent modification and allosteric effectors (see pp. 168–69).

Assimilatory Nitrate Reduction

The nitrogen in nitrate (NO_3^-) is much more oxidized than that in ammonia. Nitrate must first be reduced to ammonia before the nitrogen can be converted to an organic form. This reduction of nitrate is called **assimilatory nitrate reduction**, which is not the same as that occurring during anaerobic respiration and dissimilatory nitrate reduction (see sections 9.6 and 28.4). In assimilatory nitrate reduction, nitrate is incorporated into organic material and does not participate in energy generation. The process is widespread among bacteria, fungi, and algae.

Assimilatory nitrate reduction takes place in the cytoplasm in bacteria. The first step in nitrate assimilation is its reduction to nitrite

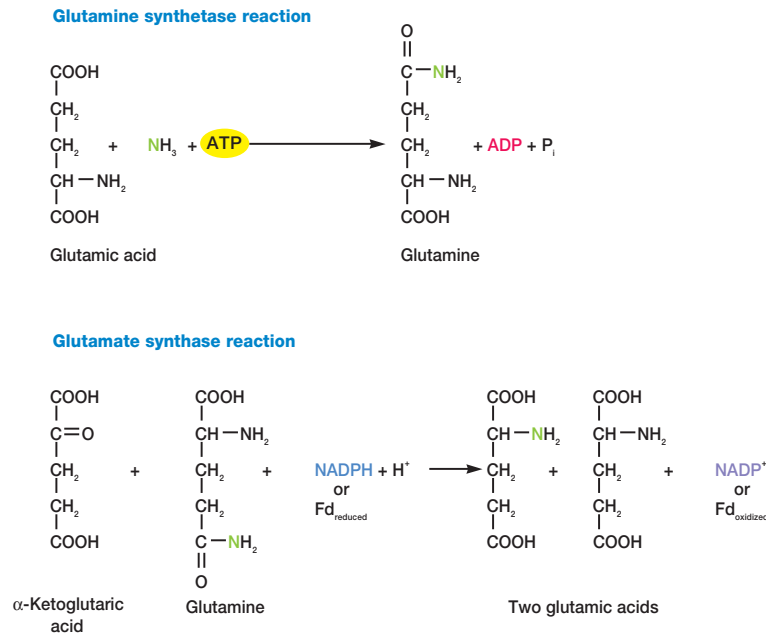


Figure 10.11 **Glutamine Synthetase and Glutamate Synthase.** The glutamine synthetase and glutamate synthase reactions involved in ammonia assimilation. Some glutamine synthetases use NADPH as an electron source; others use reduced ferredoxin (Fd). The nitrogen being incorporated and transferred is shown in green.

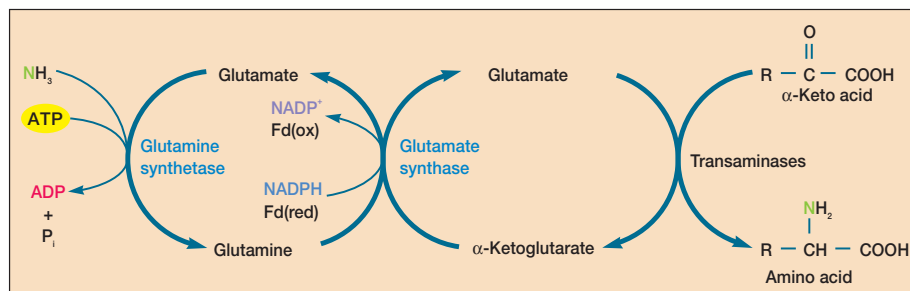


Figure 10.12 Ammonia Incorporation Using Glutamine Synthetase and Glutamate Synthase. This route is effective at low ammonia concentrations.

by **nitrate reductase**, an enzyme that contains both FAD and molybdenum (**figure 10.13**). NADPH is the electron source.



Nitrite is next reduced to ammonia with a series of two electron additions catalyzed by **nitrite reductase** and possibly other enzymes. Hydroxylamine may be an intermediate. The ammonia is then incorporated into amino acids by the routes already described.

Nitrogen Fixation

The reduction of atmospheric gaseous nitrogen to ammonia is called **nitrogen fixation**. Because ammonia and nitrate levels often are low and only a few procaryotes can carry out nitrogen fixation (eucaryotic cells completely lack this ability), the rate of this process limits plant growth in many situations. Nitrogen fixation occurs in (1) free-living bacteria (e.g., *Azotobacter*, *Klebsiella*, *Clostridium*, and *Methanococcus*), (2) bacteria living in

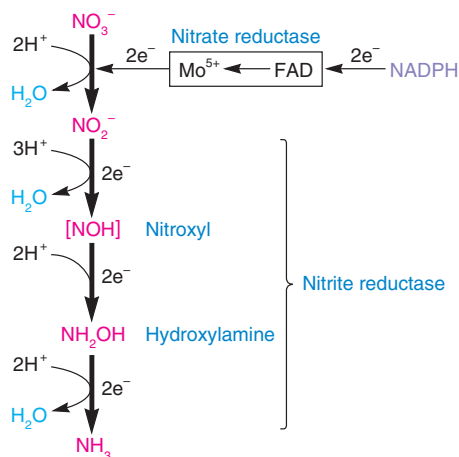


Figure 10.13 Assimilatory Nitrate Reduction. This sequence is thought to operate in bacteria that can reduce and assimilate nitrate nitrogen. See text for details.

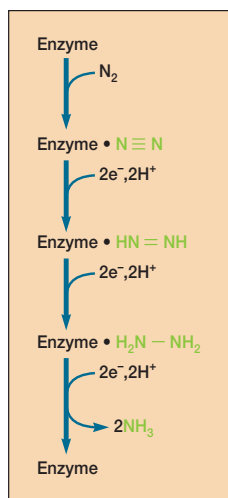


Figure 10.14 Nitrogen Reduction. A hypothetical sequence of nitrogen reduction by nitrogenase.

symbiotic association with plants such as legumes (*Rhizobium*), and (3) cyanobacteria (*Nostoc* and *Anabaena*). The biological aspects of nitrogen fixation are discussed in chapter 30. The biochemistry of nitrogen fixation is the focus of this section. [The biology of nitrogen-fixing microorganisms](#) (pp. 492, 616, 675–78)

The reduction of nitrogen to ammonia is catalyzed by the enzyme **nitrogenase**. Although the enzyme-bound intermediates in this process are still unknown, it is believed that nitrogen is reduced by two-electron additions in a way similar to that illustrated in **figure 10.14**. The reduction of molecular nitrogen to ammonia is

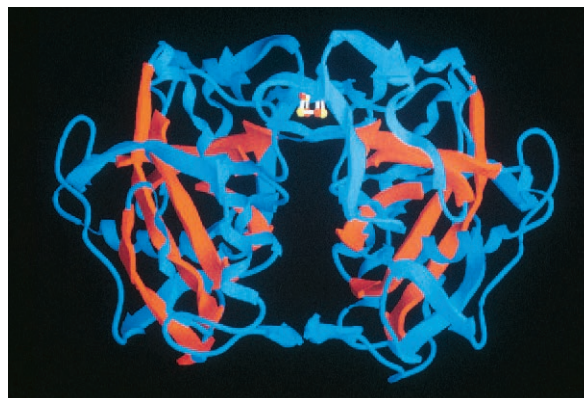
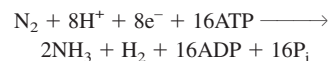


Figure 10.15 Structure of the Nitrogenase Fe Protein. The Fe protein's two subunits are arranged like a pair of butterfly wings with the iron sulfur cluster between the wings and at the "head" of the butterfly. The iron sulfur cluster is very exposed, which helps account for nitrogenase's sensitivity to oxygen. The oxygen can readily attack the exposed irons.

quite exergonic, but the reaction has a high activation energy because molecular nitrogen is an unreactive gas with a triple bond between the two nitrogen atoms. Therefore nitrogen reduction is expensive and requires a large ATP expenditure. At least 8 electrons and 16 ATP molecules, 4 ATPs per pair of electrons, are required.



The electrons come from ferredoxin that has been reduced in a variety of ways: by photosynthesis in cyanobacteria, respiratory processes in aerobic nitrogen fixers, or fermentations in anaerobic bacteria. For example, *Clostridium pasteurianum* (an anaerobic bacterium) reduces ferredoxin during pyruvate oxidation, whereas the aerobic *Azotobacter* uses electrons from NADPH to reduce ferredoxin.

Nitrogenase is a complex system consisting of two major protein components, a MoFe protein (MW 220,000) joined with one or two Fe proteins (MW 64,000). The MoFe protein contains 2 atoms of molybdenum and 28 to 32 atoms of iron; the Fe protein has 4 iron atoms (**figure 10.15**). Fe protein is first reduced by ferredoxin, then it binds ATP (**figure 10.16**). ATP binding changes the conformation of the Fe protein and lowers its reduction potential, enabling it to reduce the MoFe protein. ATP is hydrolyzed when this electron transfer occurs. Finally, reduced MoFe protein donates electrons to atomic nitrogen. Nitrogenase is quite sensitive to O_2 and must be protected from O_2 inactivation within the cell. In many cyanobacteria, this protection against oxygen is provided by a special structure called the heterocyst (*see p. 473*).

The reduction of N_2 to NH_3 occurs in three steps, each of which requires an electron pair (figures 10.14 and 10.16). Six electron transfers take place, and this requires a total 12 ATPs per N_2 reduced. The overall process actually requires at least 8 electrons